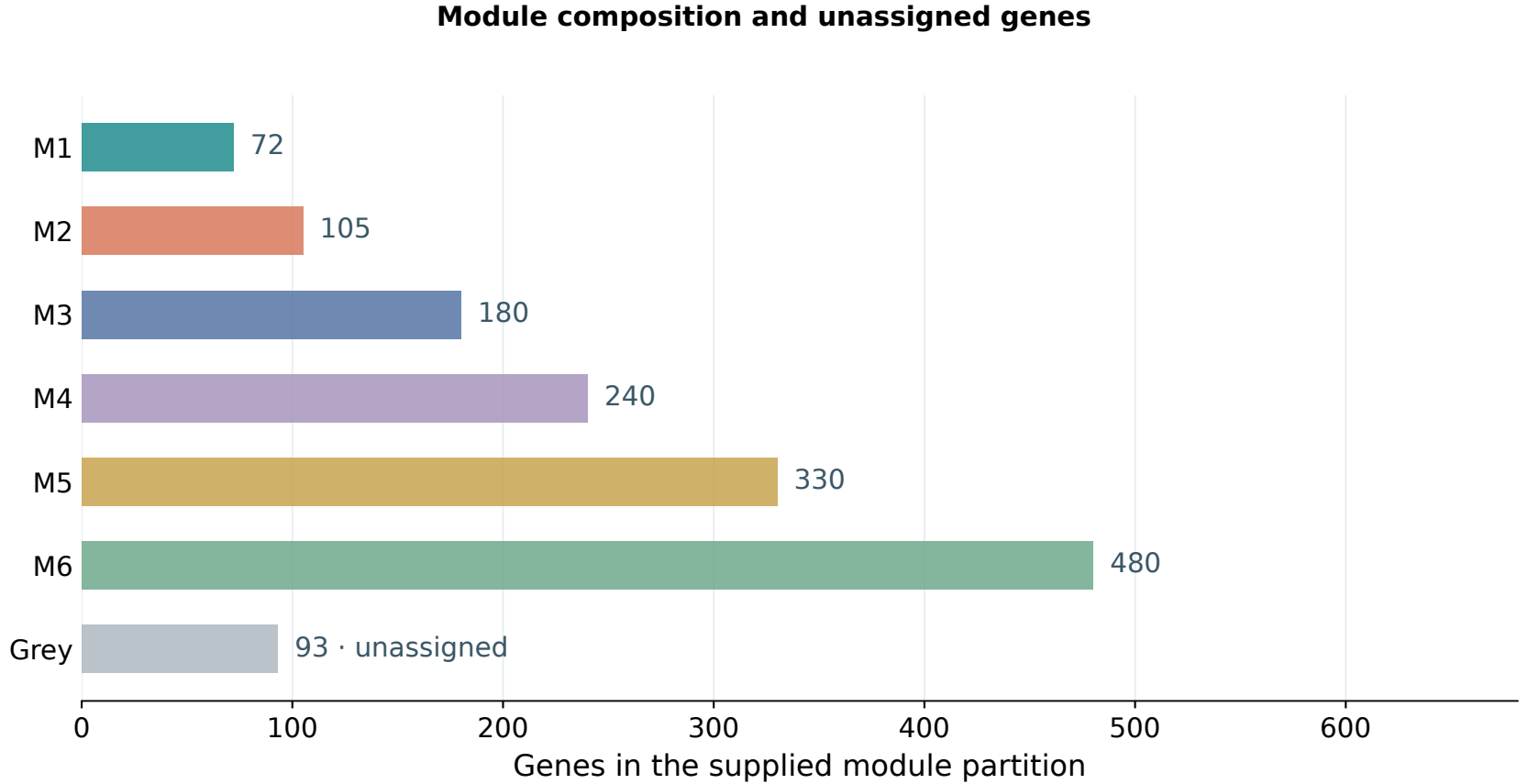
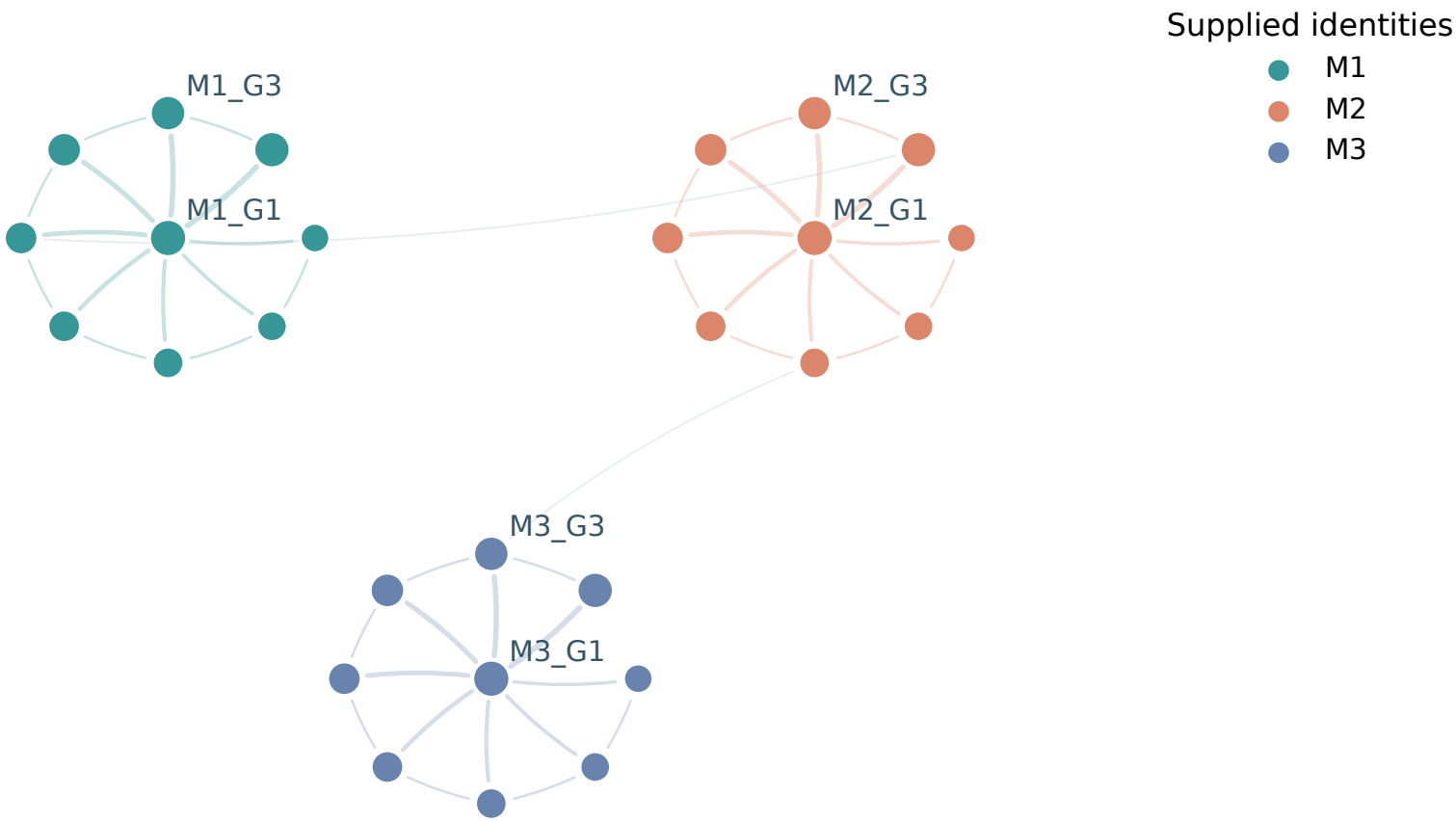
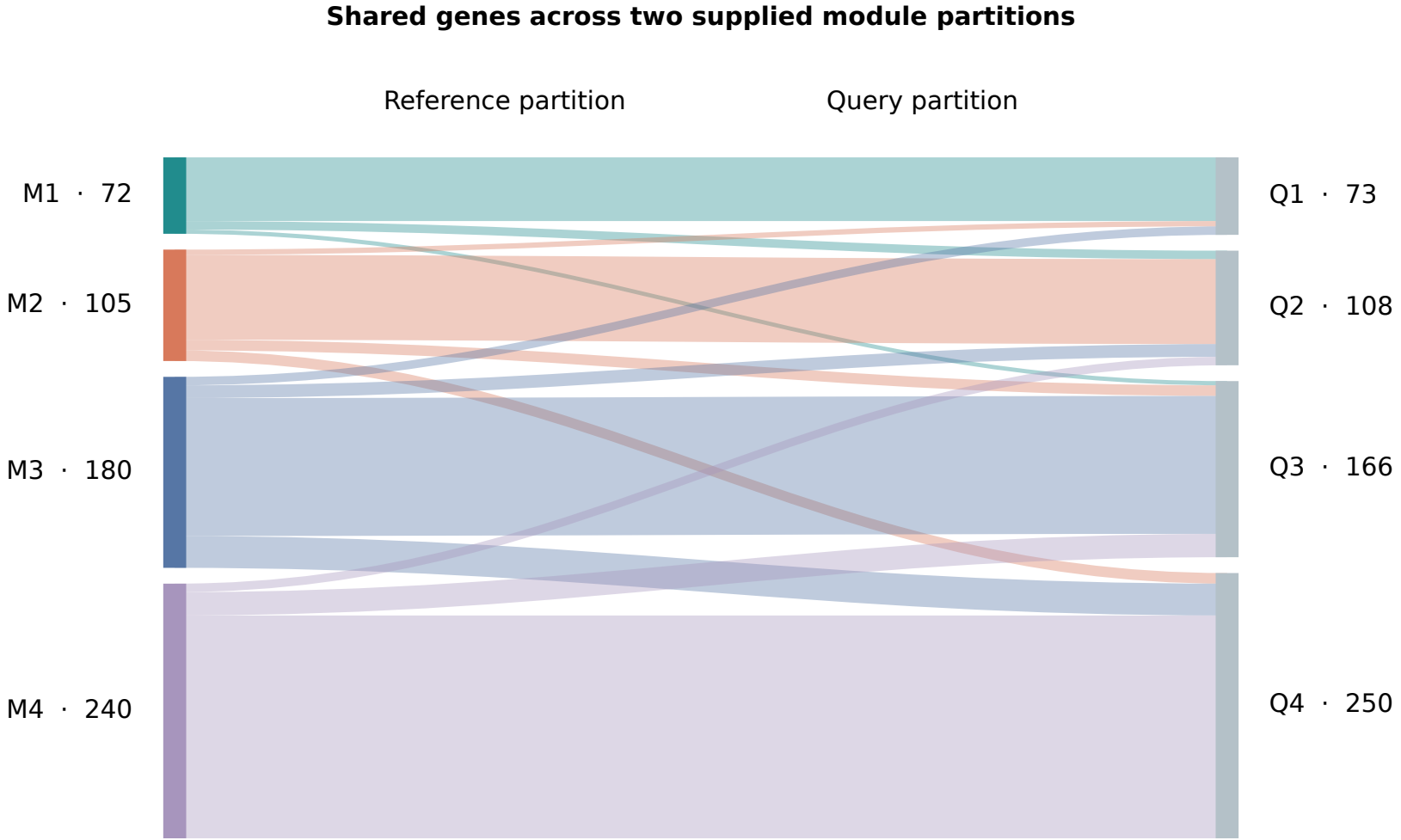

Module composition and unassigned genes
M1
72
M2
105
M3
180
M4
240
M5
330
M6
480
Grey
93 · unassigned
0
100
200
300
400
500
600
Genes in the supplied module partition

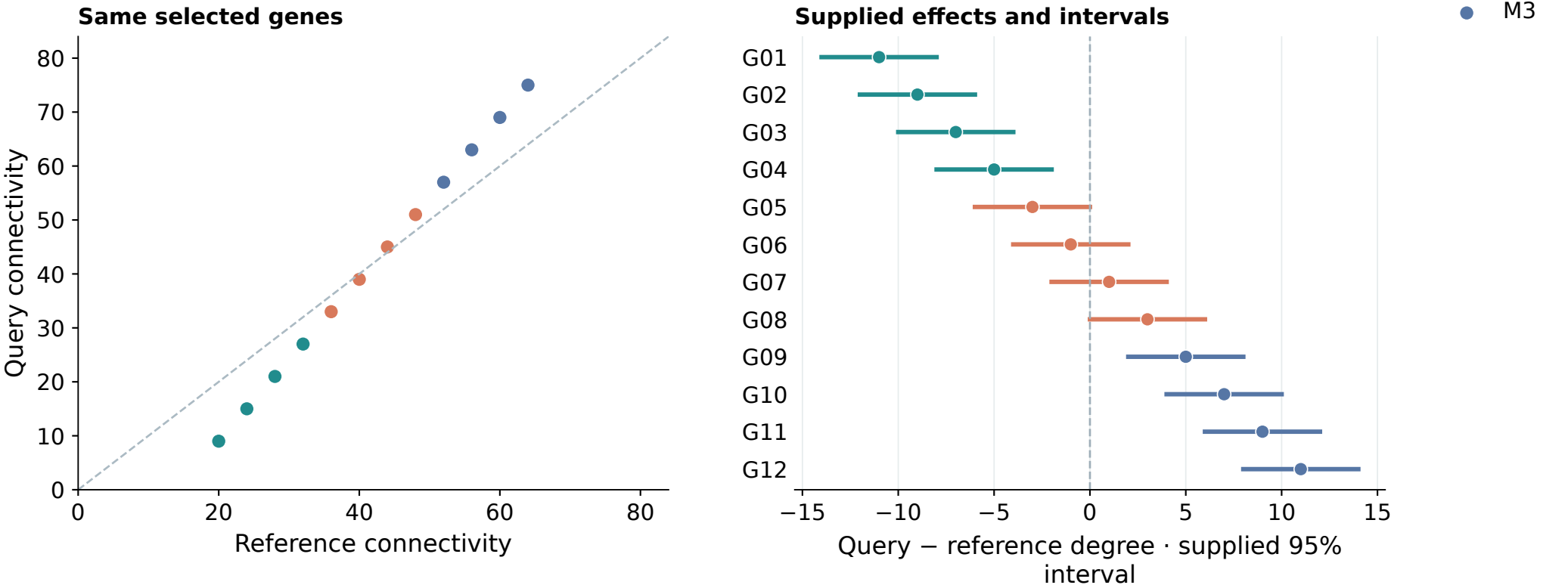
Coexpression neighborhoods and hub candidates

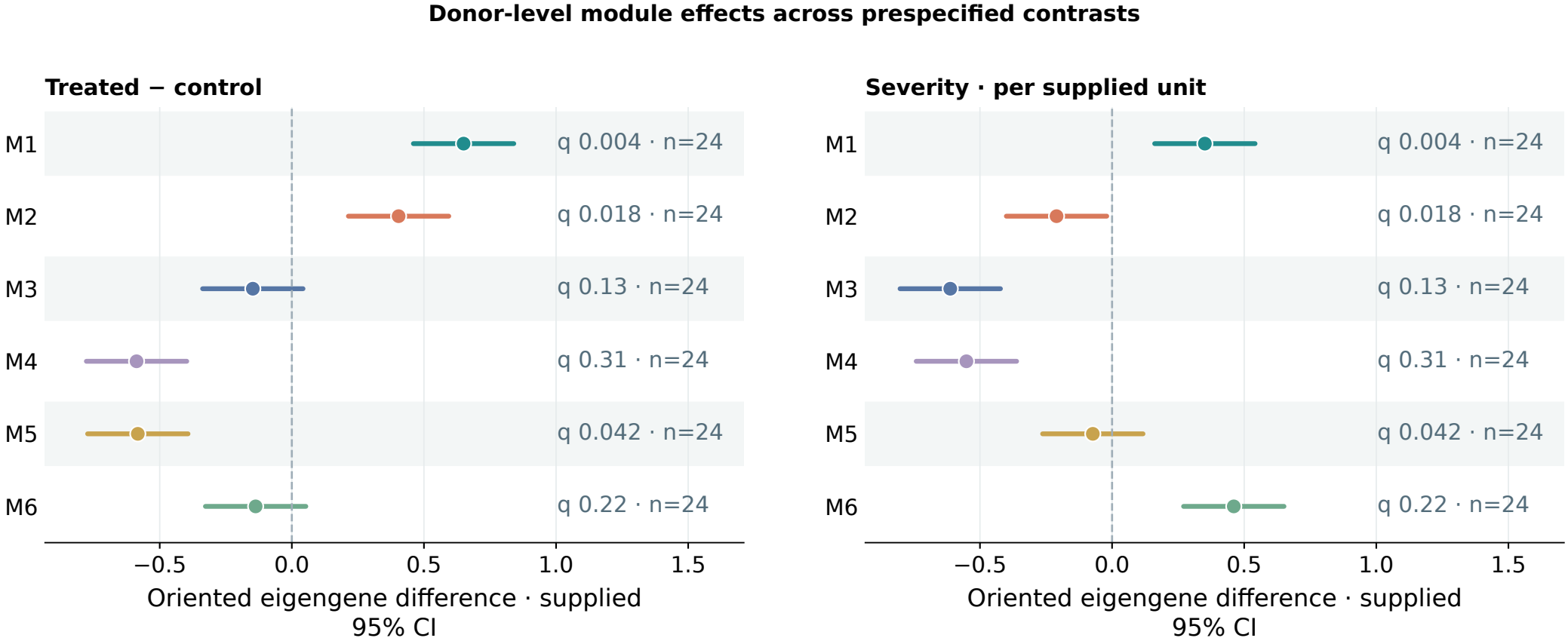


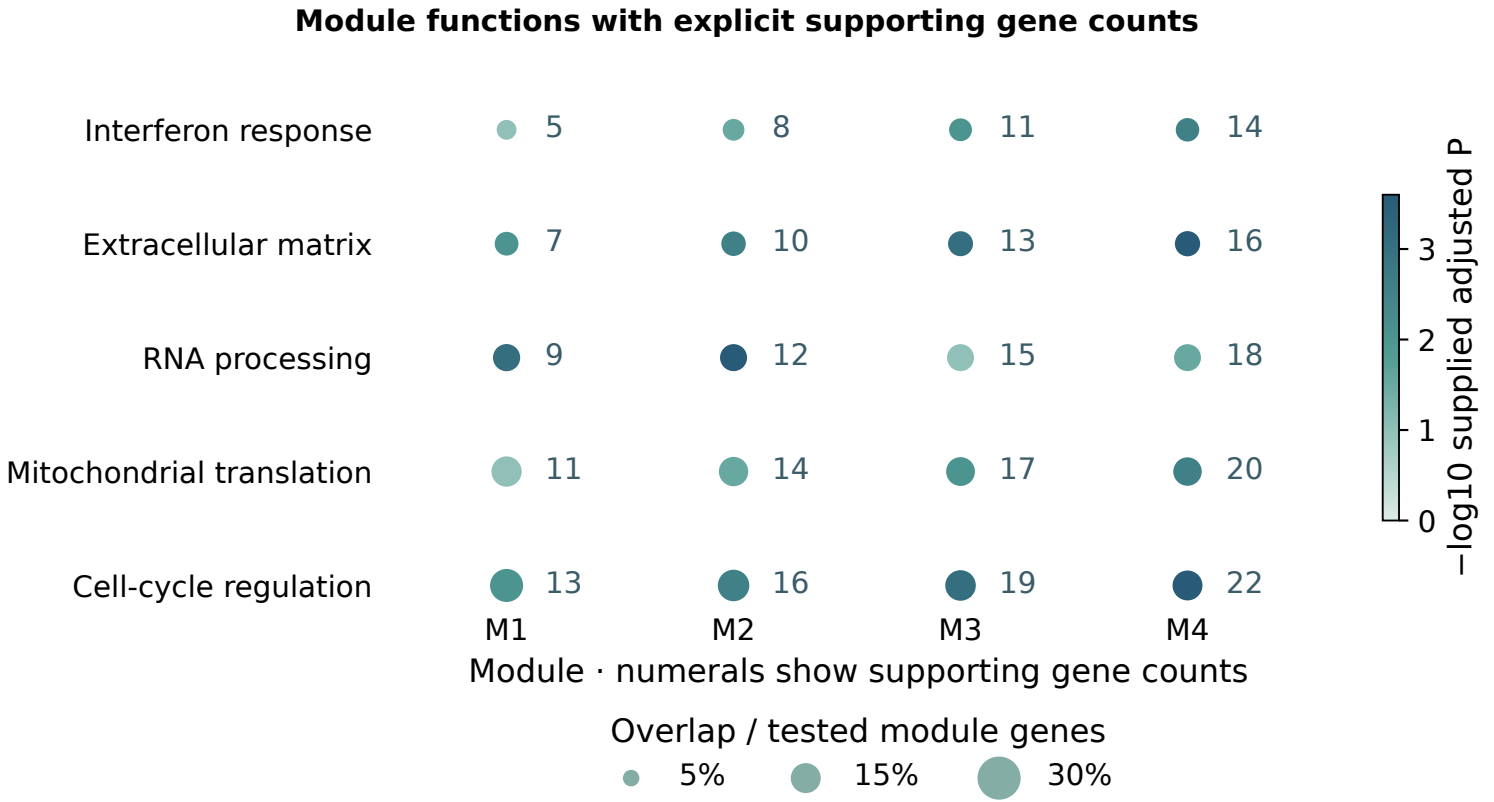
27 selected nodes · 47 supplied edges · point area scaled by |kME|


Shared genes across two supplied module partitions
Reference partition
Query partition
M1 · 72
M2 · 105
M3 · 180
M4 · 240
Q1 · 73
Q2 · 108
Q3 · 166
Q4 · 250

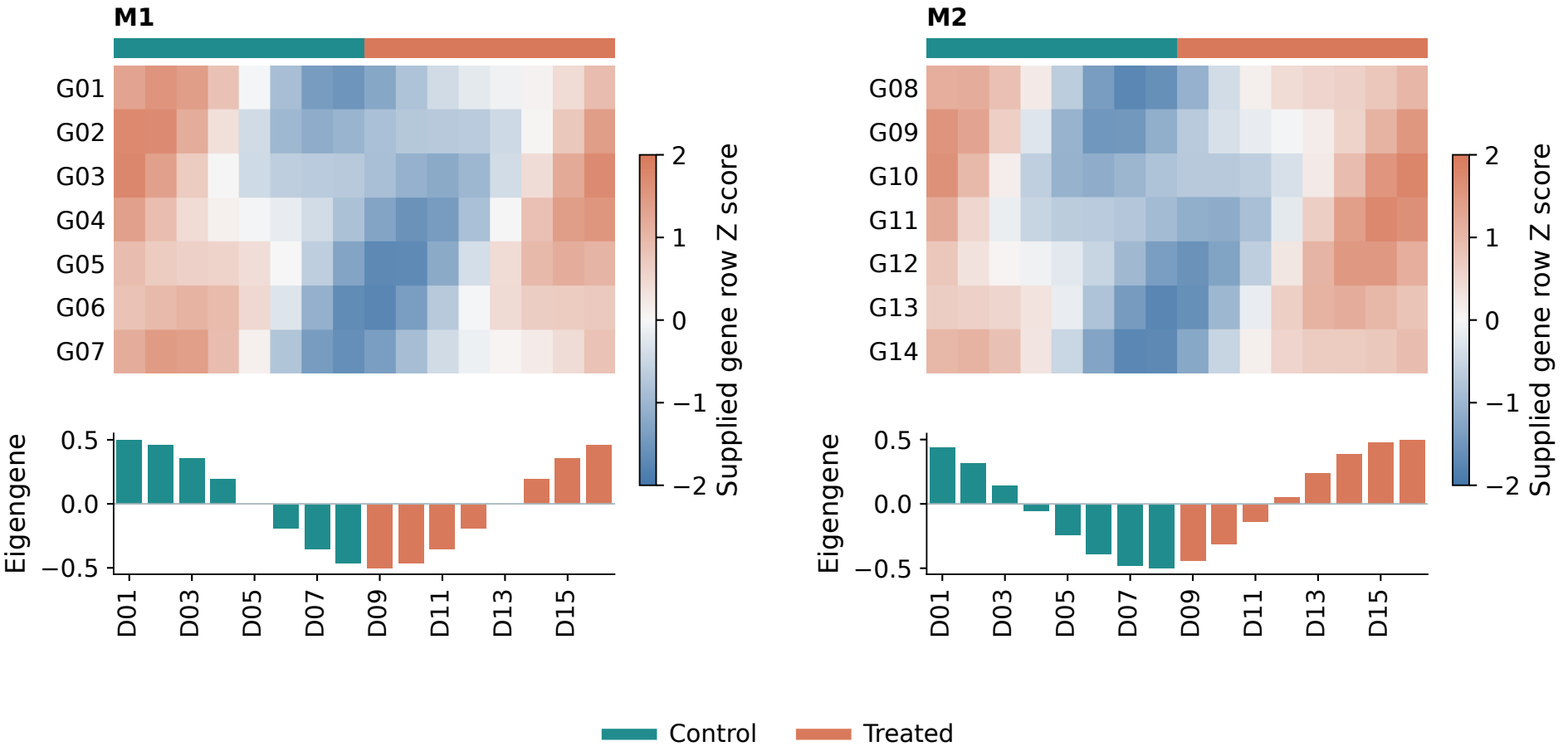
Connectivity change on identical selected genes



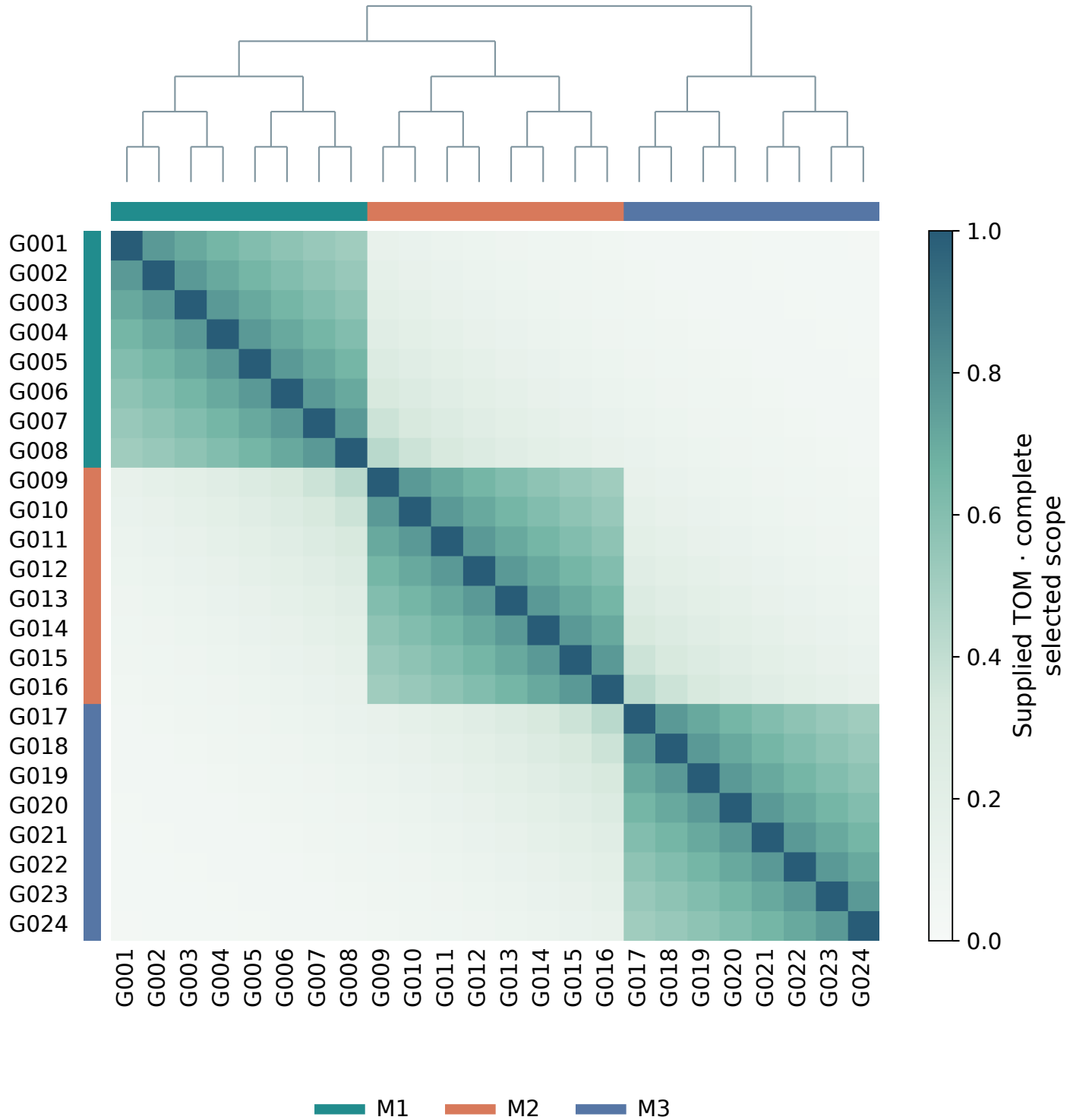

Donor-level module effects across prespecified contrasts
Treated − control
M1
M2
M3
M4
M5
M6
q 0.004 · n=24
q 0.018 · n=24
q 0.13 · n=24
q 0.31 · n=24
q 0.042 · n=24
q 0.22 · n=24
−0.5
0.0
0.5
1.0
1.5
Oriented eigengene difference · supplied
95% CI
Severity · per supplied unit
M1
M2
M3
M4
M5
M6
q 0.004 · n=24
q 0.018 · n=24
q 0.13 · n=24
q 0.31 · n=24
q 0.042 · n=24
q 0.22 · n=24
−0.5
0.0
0.5
1.0
1.5
Oriented eigengene difference · supplied
95% CI


Module functions with explicit supporting gene counts
Interferon response
Extracellular matrix
RNA processing
Mitochondrial translation
Cell-cycle regulation
5
8
11
14
7
10
13
16
9
12
15
18
11
14
17
20
13
16
19
22
M1
M2
M3
M4
Module · numerals show supporting gene counts
Overlap / tested module genes
5%
15%
30%
−log10 supplied adjusted P
0
1
2
3

Selected module genes aligned with donor eigengenes

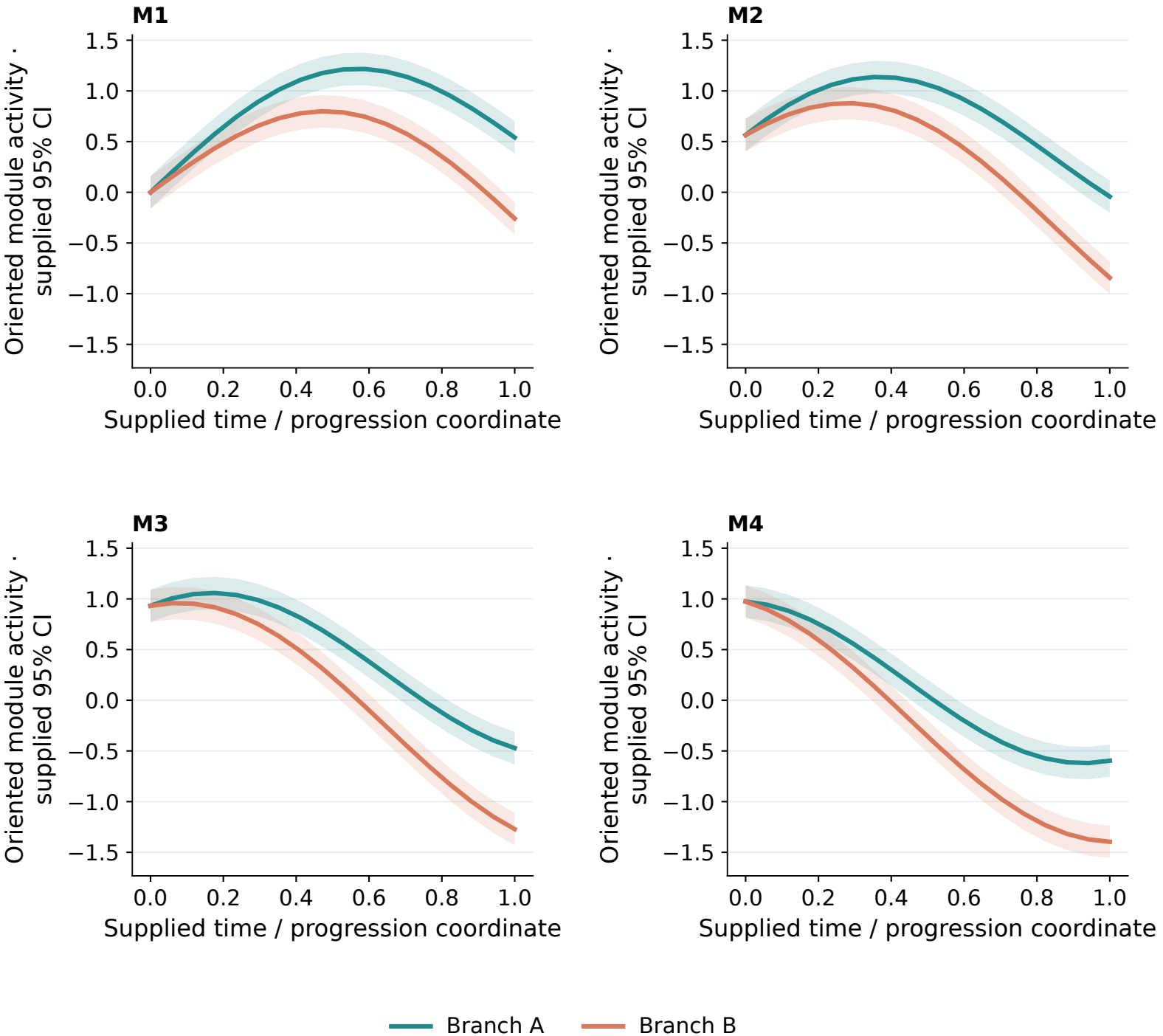


Topological overlap with frozen gene hierarchy and modules

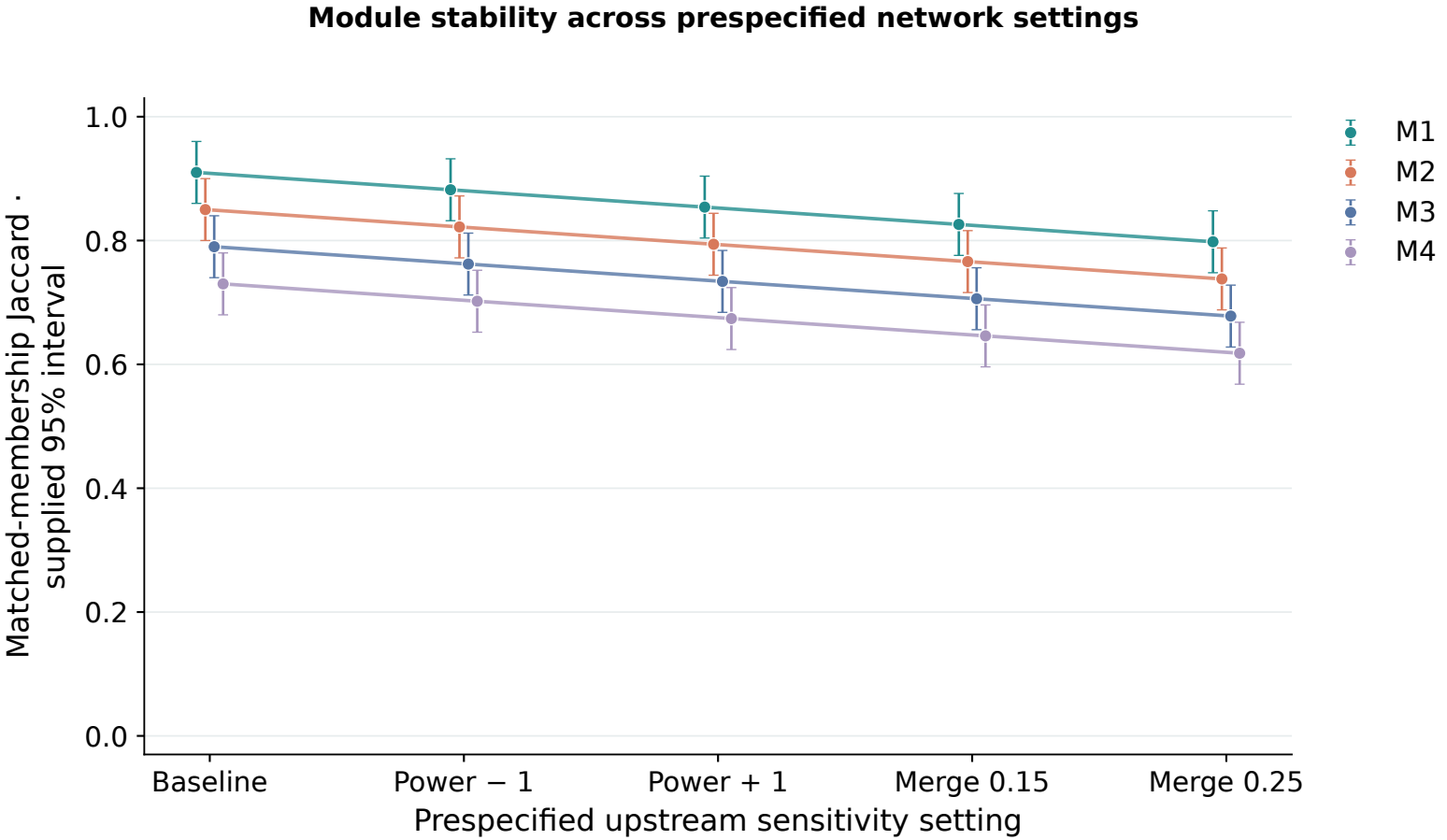


Software test fixture; not biological evidence

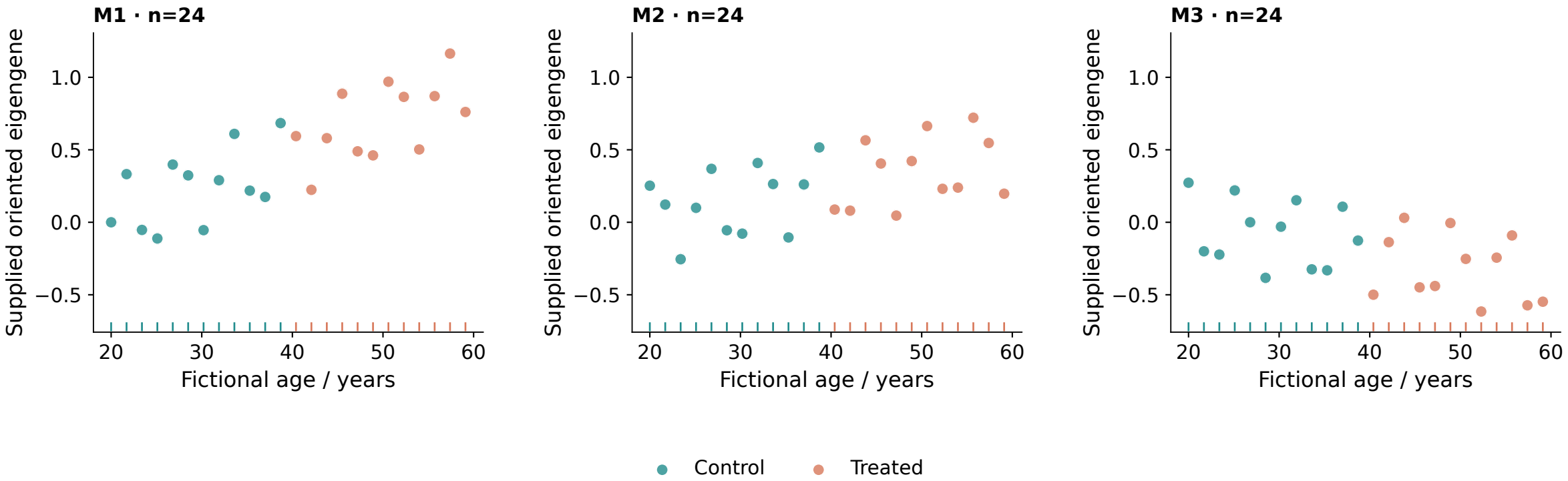
Module programs along supplied branches



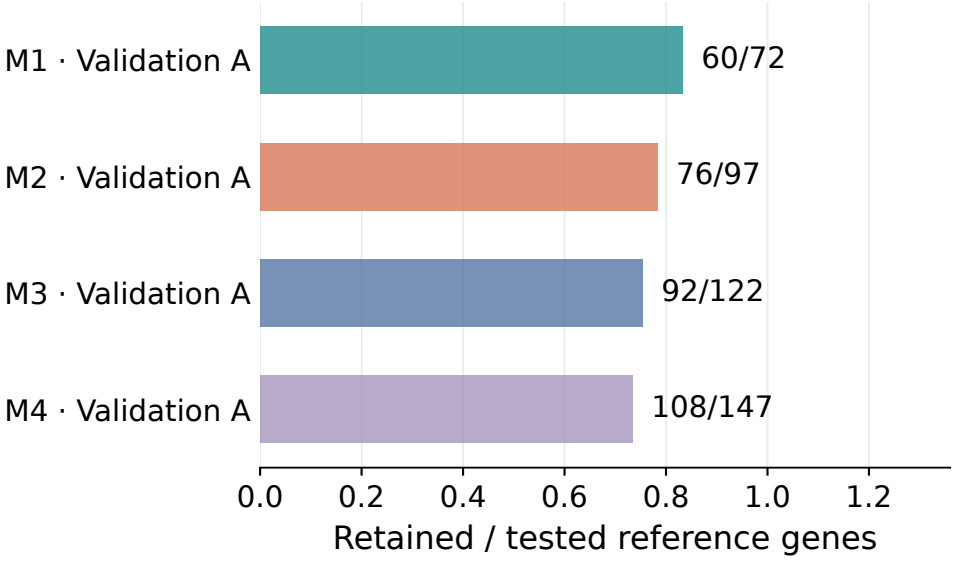
Software test fixture; not biological evidence

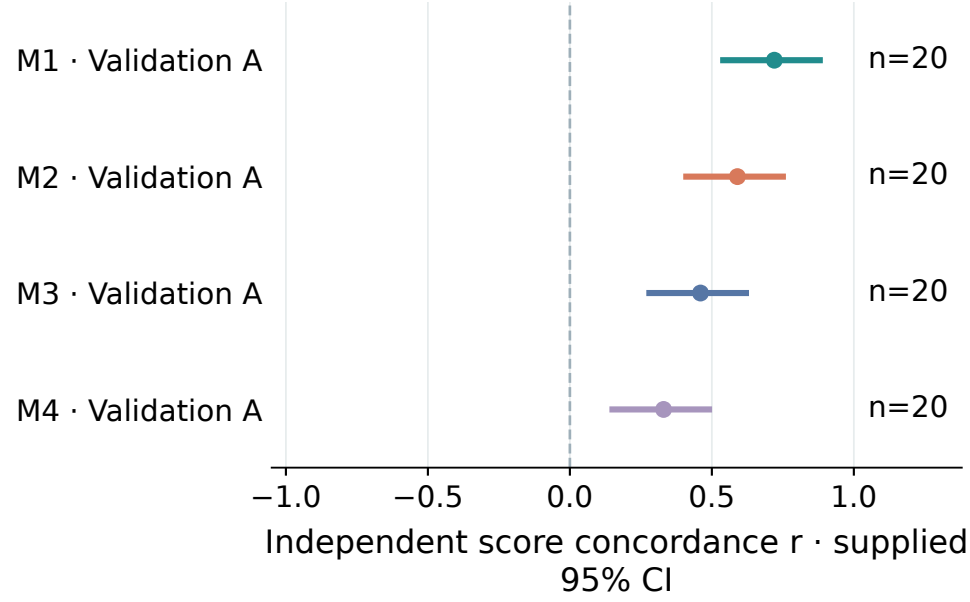

Module stability across prespecified network settings
Matched-membership Jaccard · supplied 95% interval
1.0
0.8
0.6
0.4
0.2
0.0
Baseline
Power − 1
Power + 1
Merge 0.15
Merge 0.25
Prespecified upstream sensitivity setting
M1
M2
M3
M4

Continuous trait relationships at the donor level

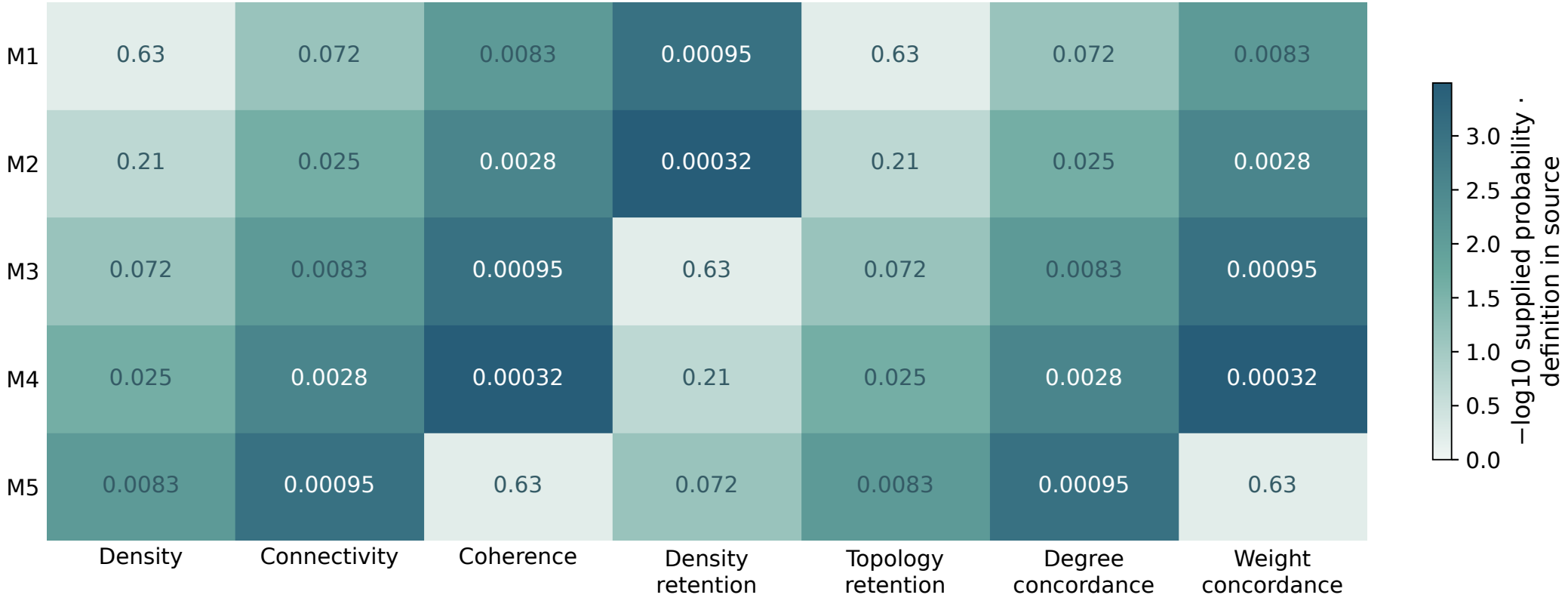


External module projection: coverage and concordance

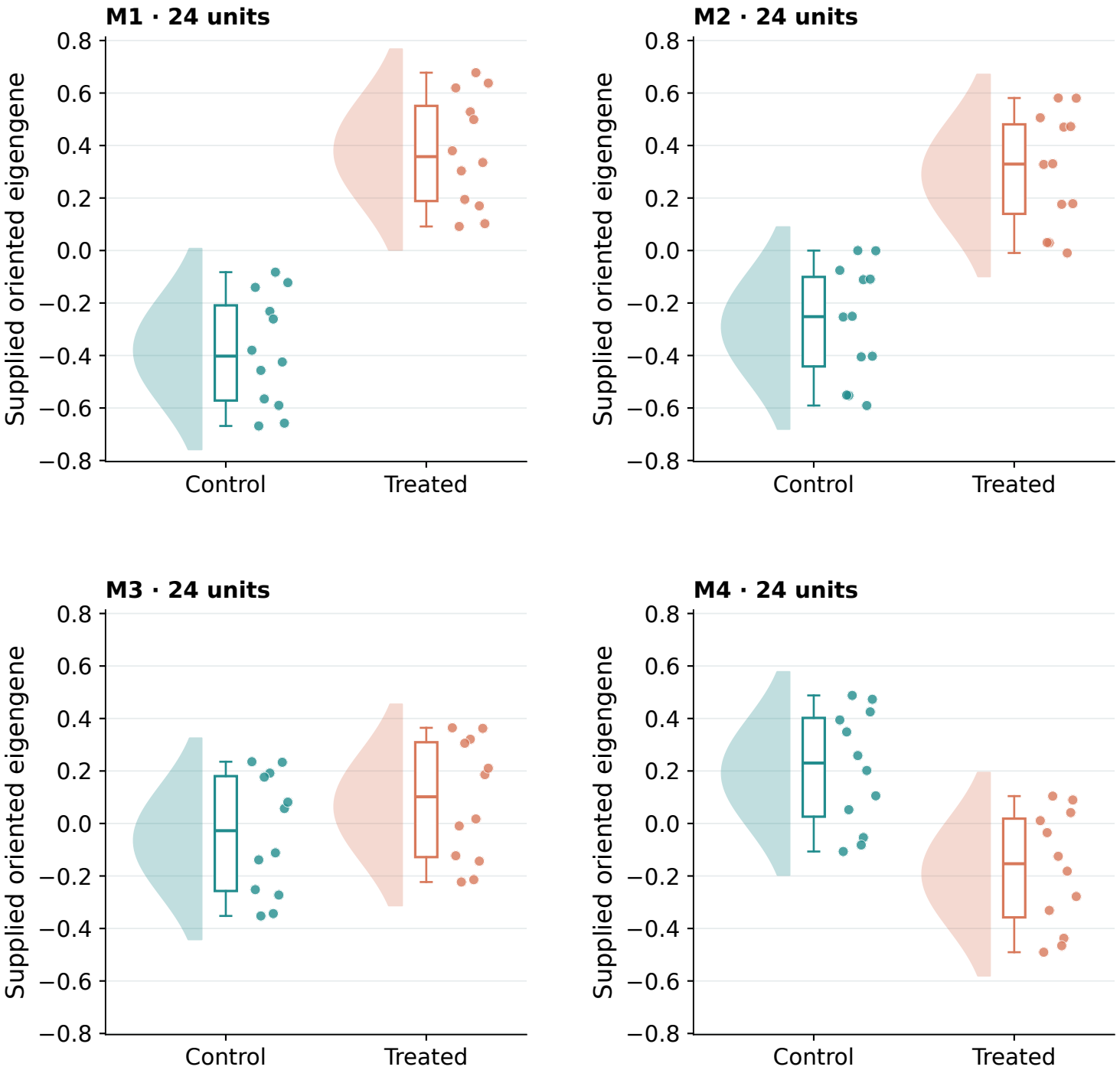




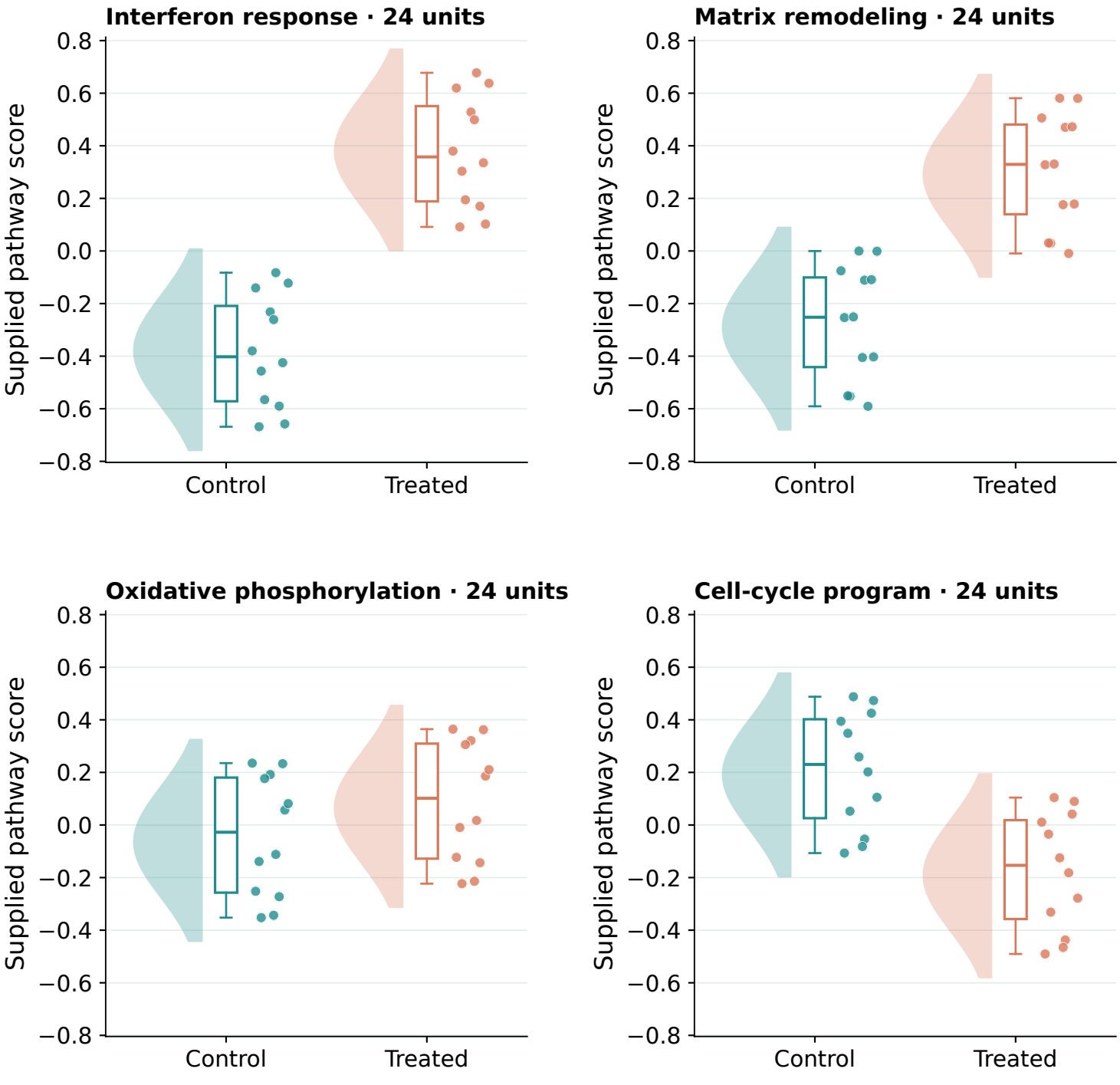
Preservation evidence for each network attribute



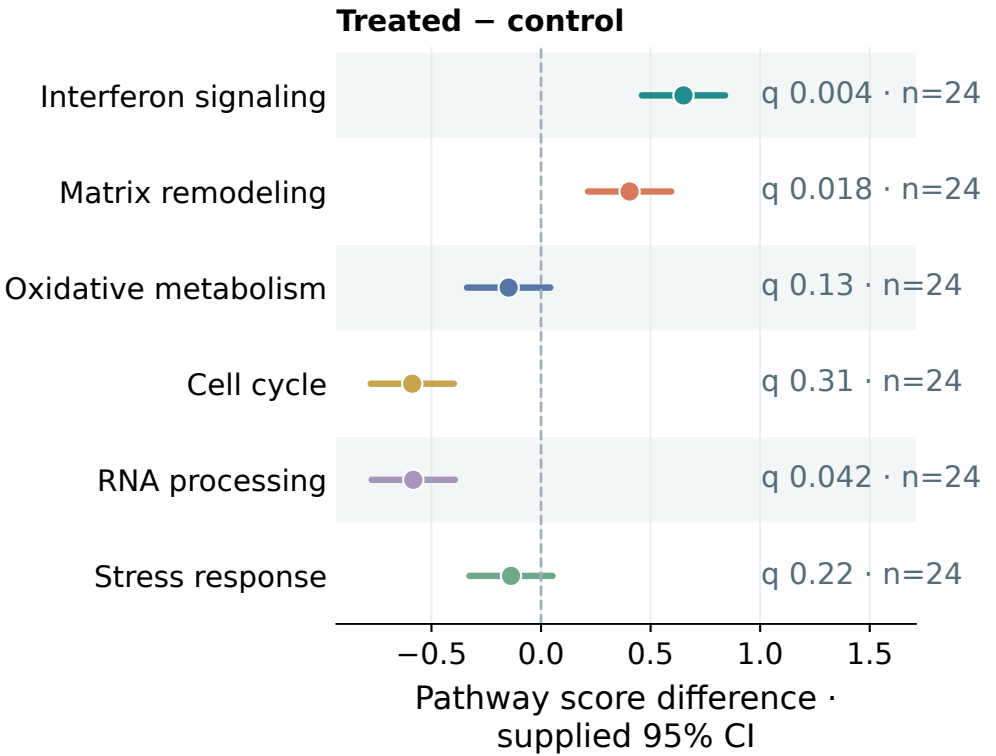
Donor-level eigengenes: distributions and individual units

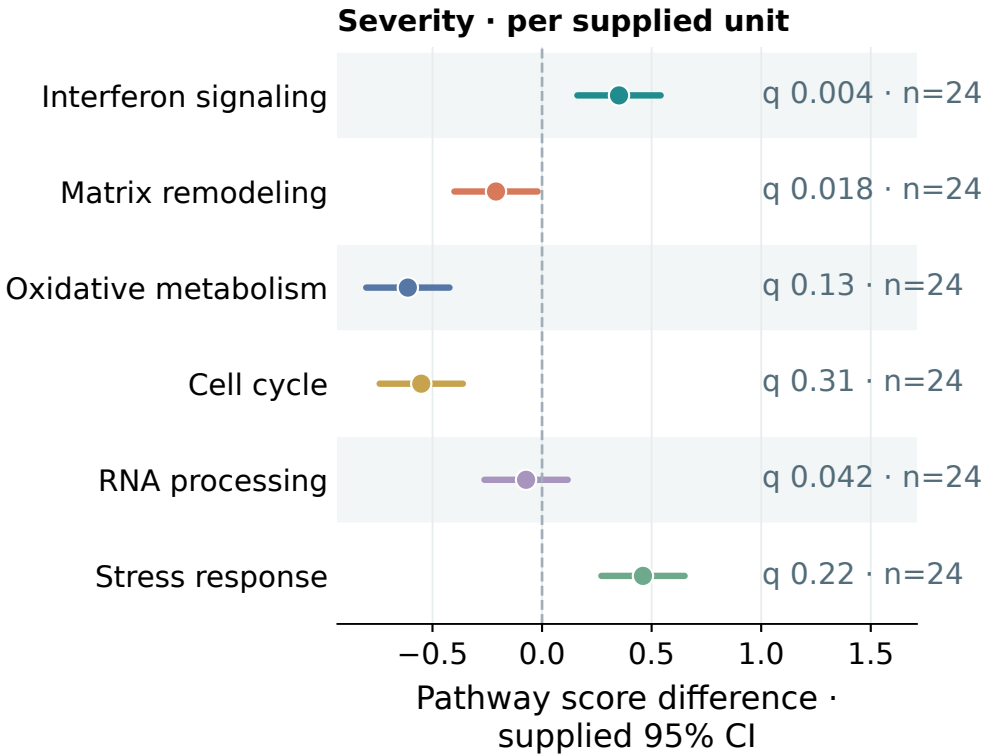


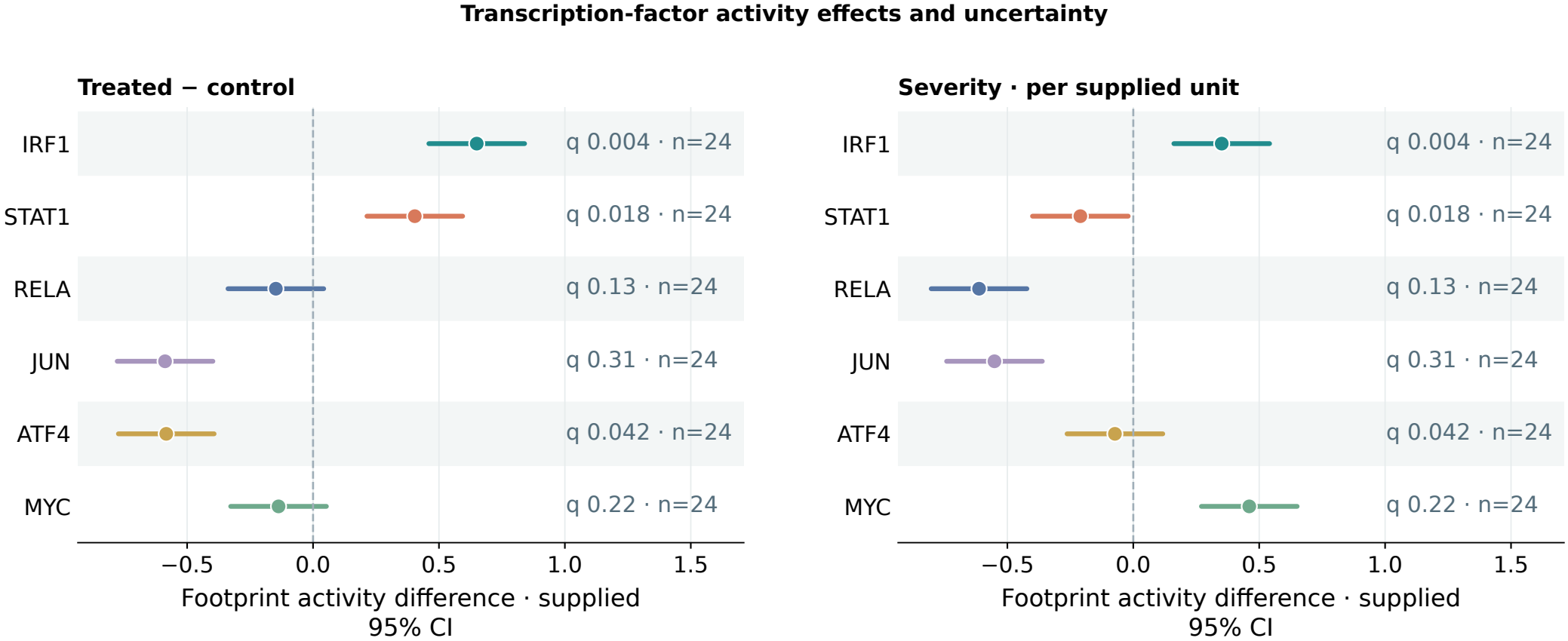
Pathway score distributions at the donor level



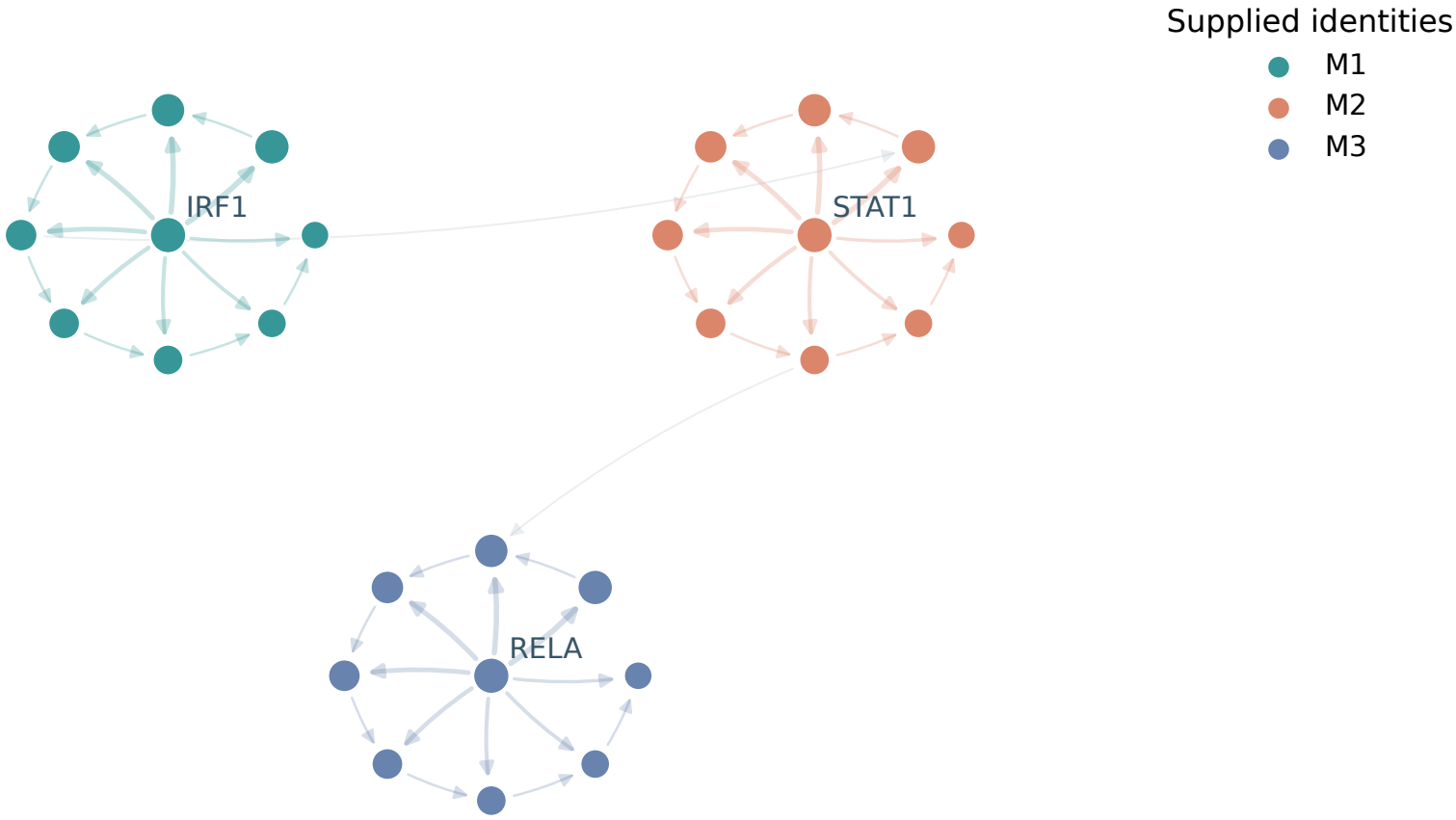
Adjusted pathway score effects across contrasts






Transcription-factor activity effects and uncertainty
Treated − control
IRF1
STAT1
RELA
JUN
ATF4
MYC
q 0.004 · n=24
q 0.018 · n=24
q 0.13 · n=24
q 0.31 · n=24
q 0.042 · n=24
q 0.22 · n=24
−0.5
0.0
0.5
1.0
1.5
Footprint activity difference · supplied
95% CI
Severity · per supplied unit
IRF1
STAT1
RELA
JUN
ATF4
MYC
q 0.004 · n=24
q 0.018 · n=24
q 0.13 · n=24
q 0.31 · n=24
q 0.042 · n=24
q 0.22 · n=24
−0.5
0.0
0.5
1.0
1.5
Footprint activity difference · supplied
95% CI

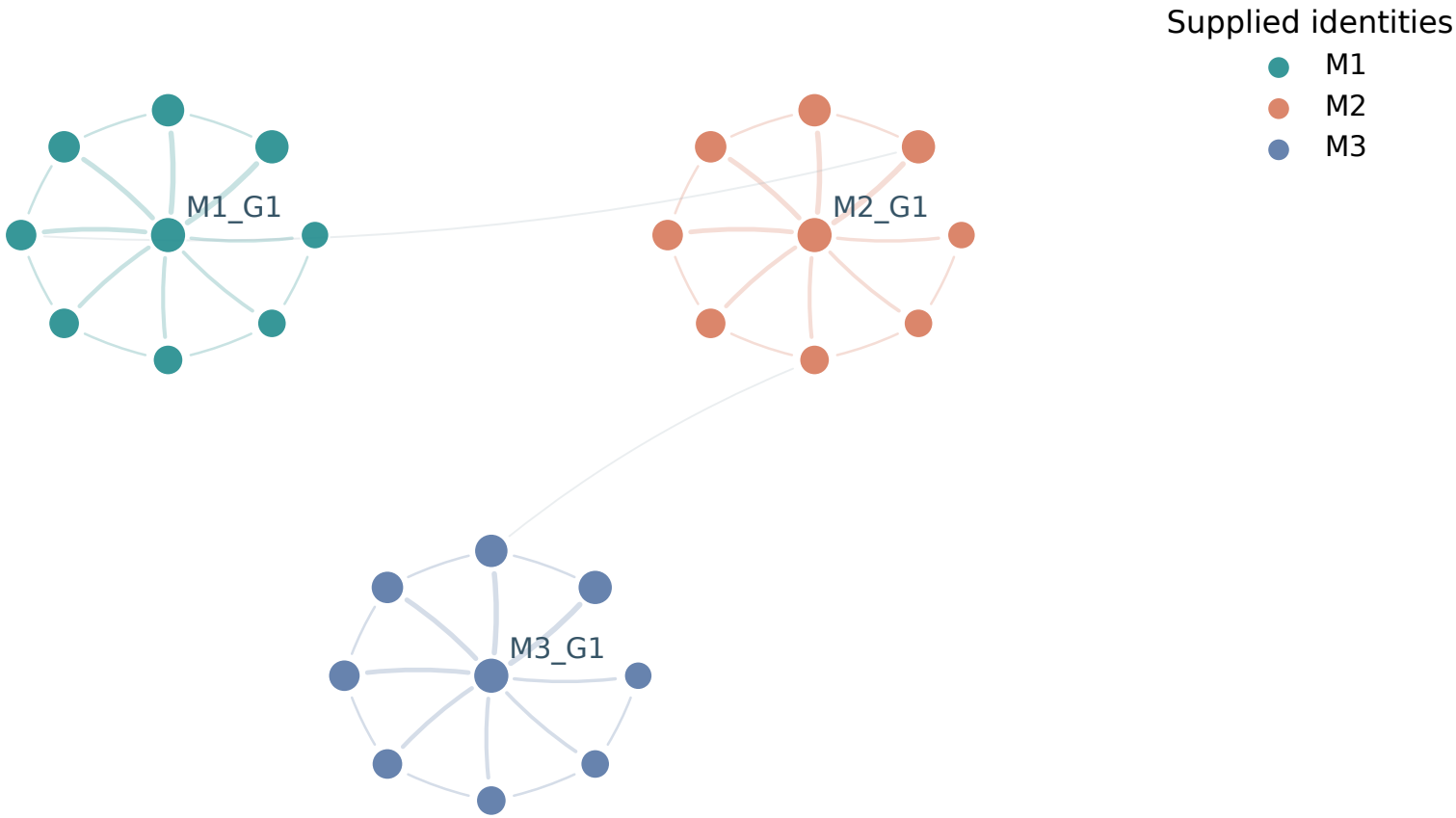
Regulon target support with explicit edge direction



27 selected nodes · 47 supplied edges · point area scaled by |kME|

Software test fixture; not biological evidence

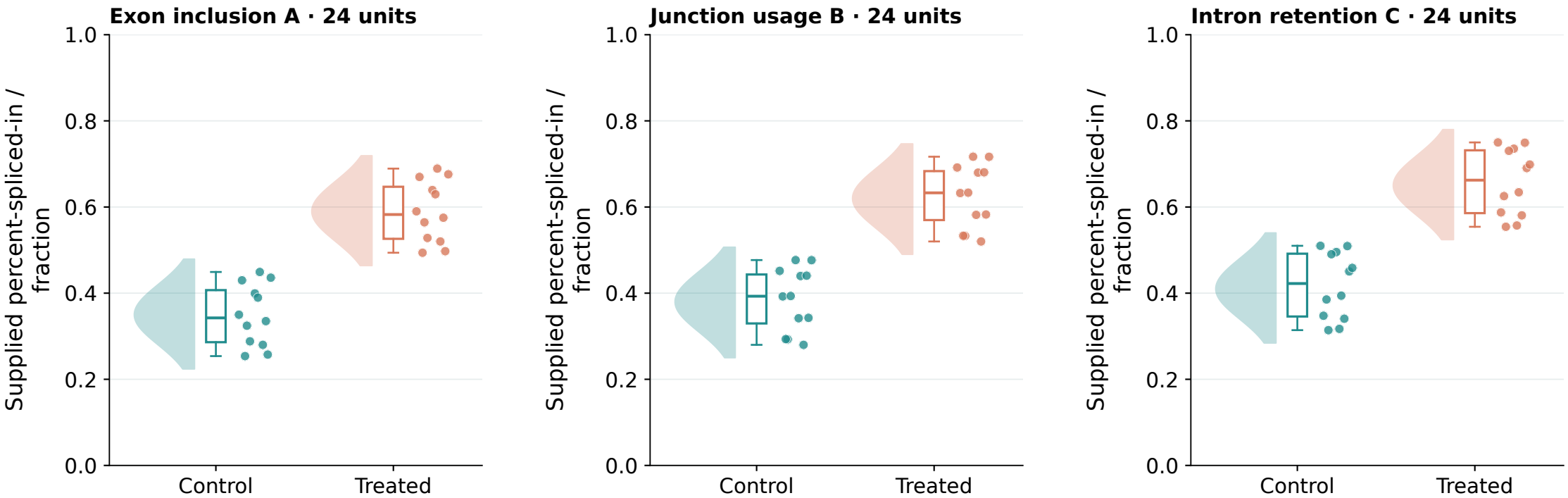
Selected protein interactions and module context



27 selected nodes · 47 supplied edges · point area scaled by |kME|

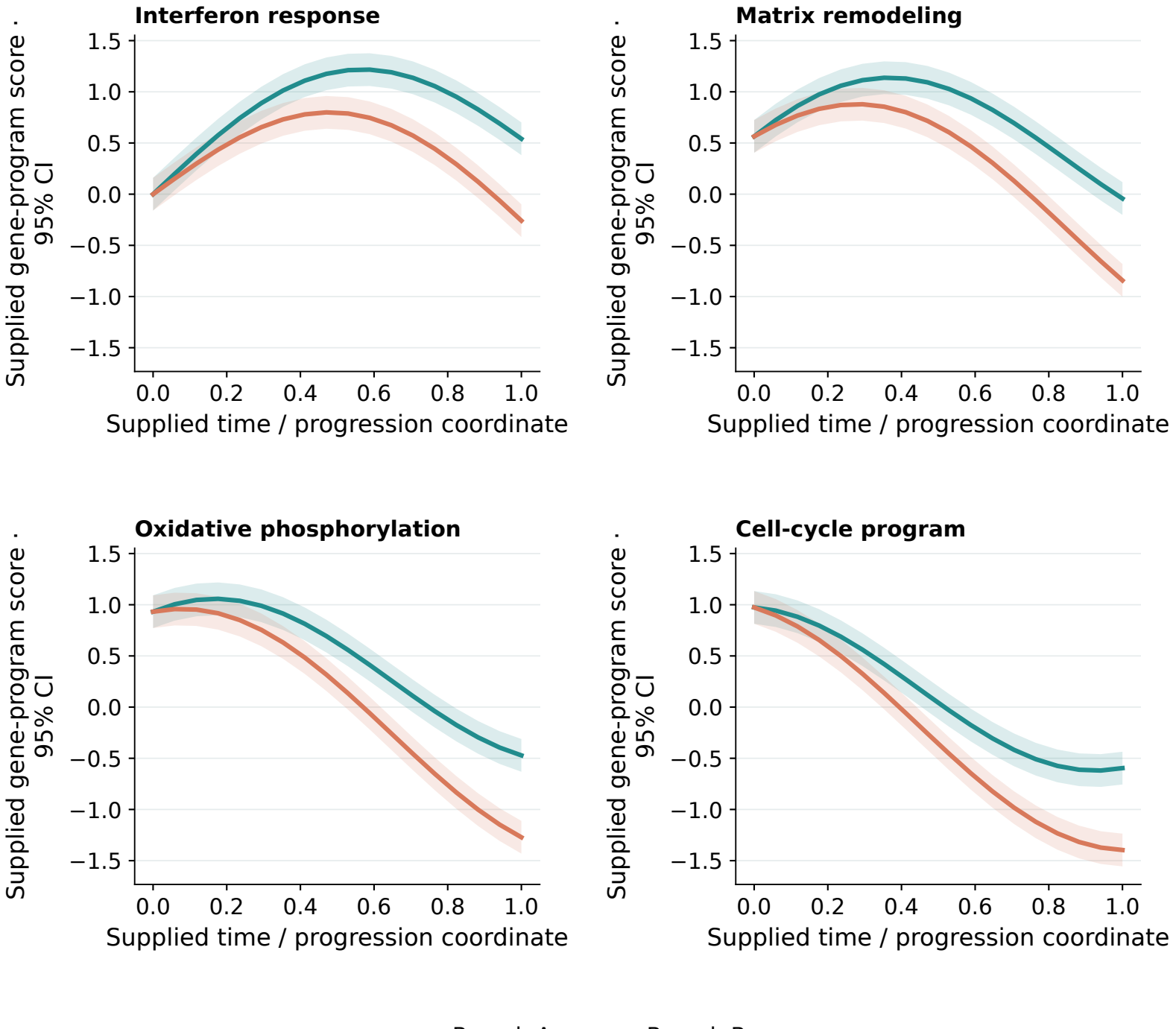
Software test fixture; not biological evidence

Splicing events: donor-level inclusion distributions

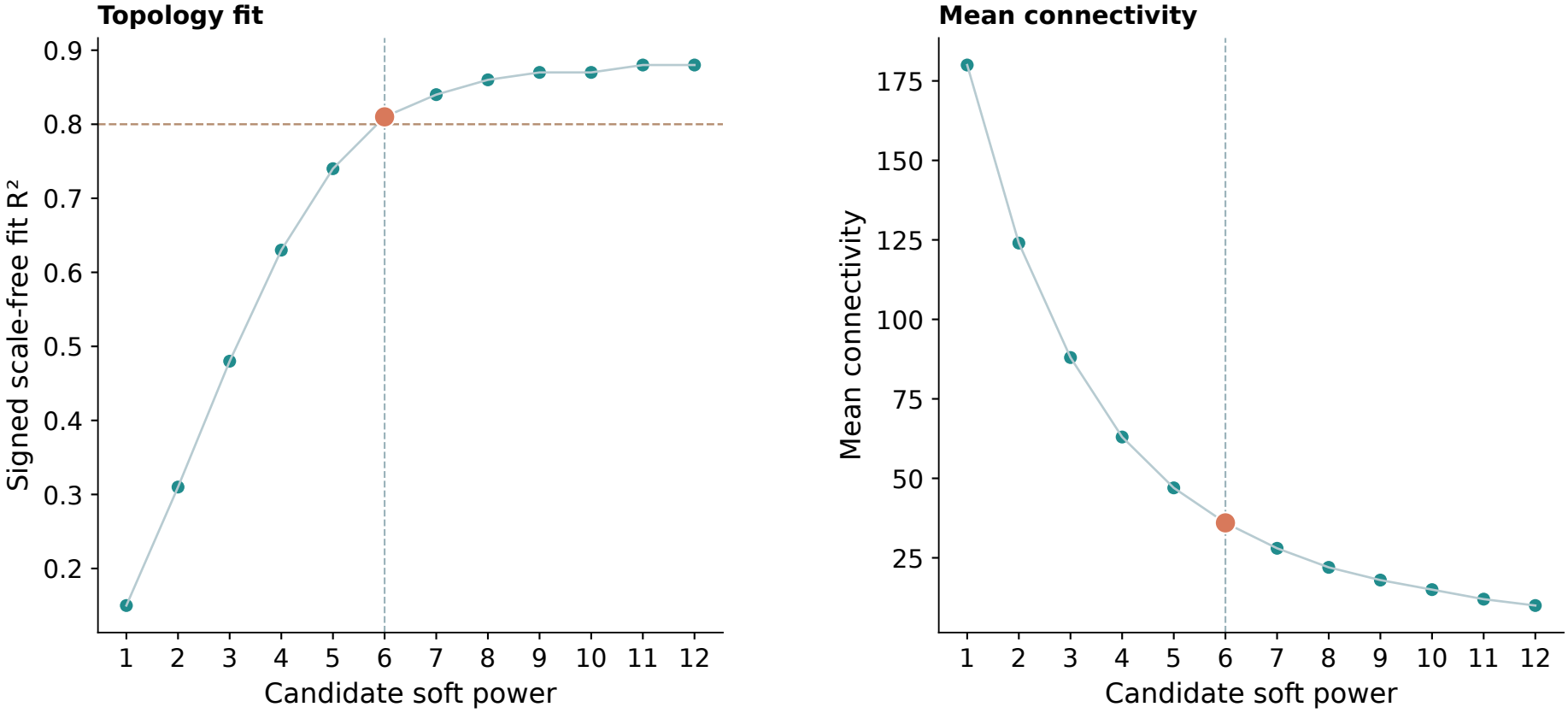


Software test fixture; not biological evidence

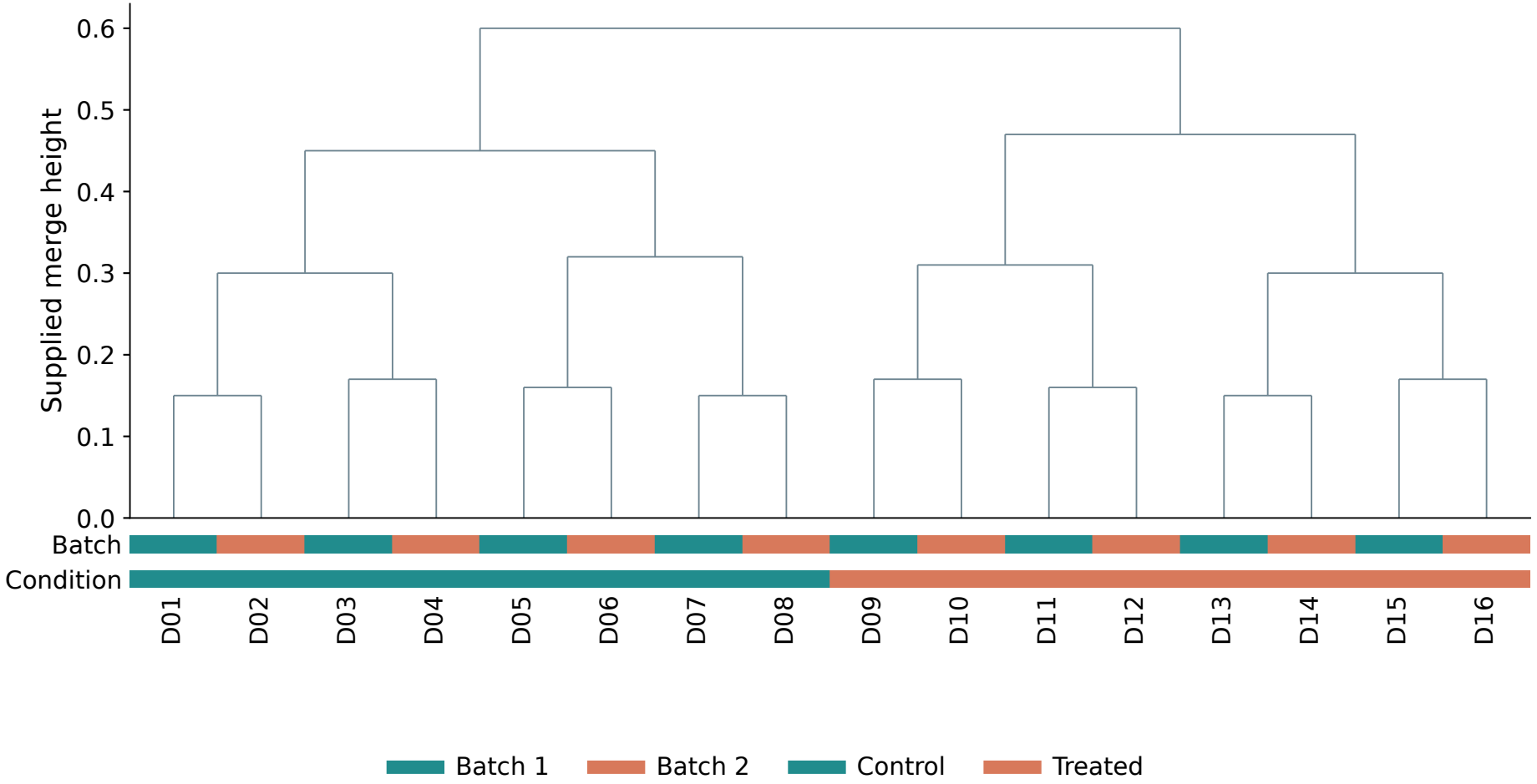
Gene programs along a supplied progression coordinate

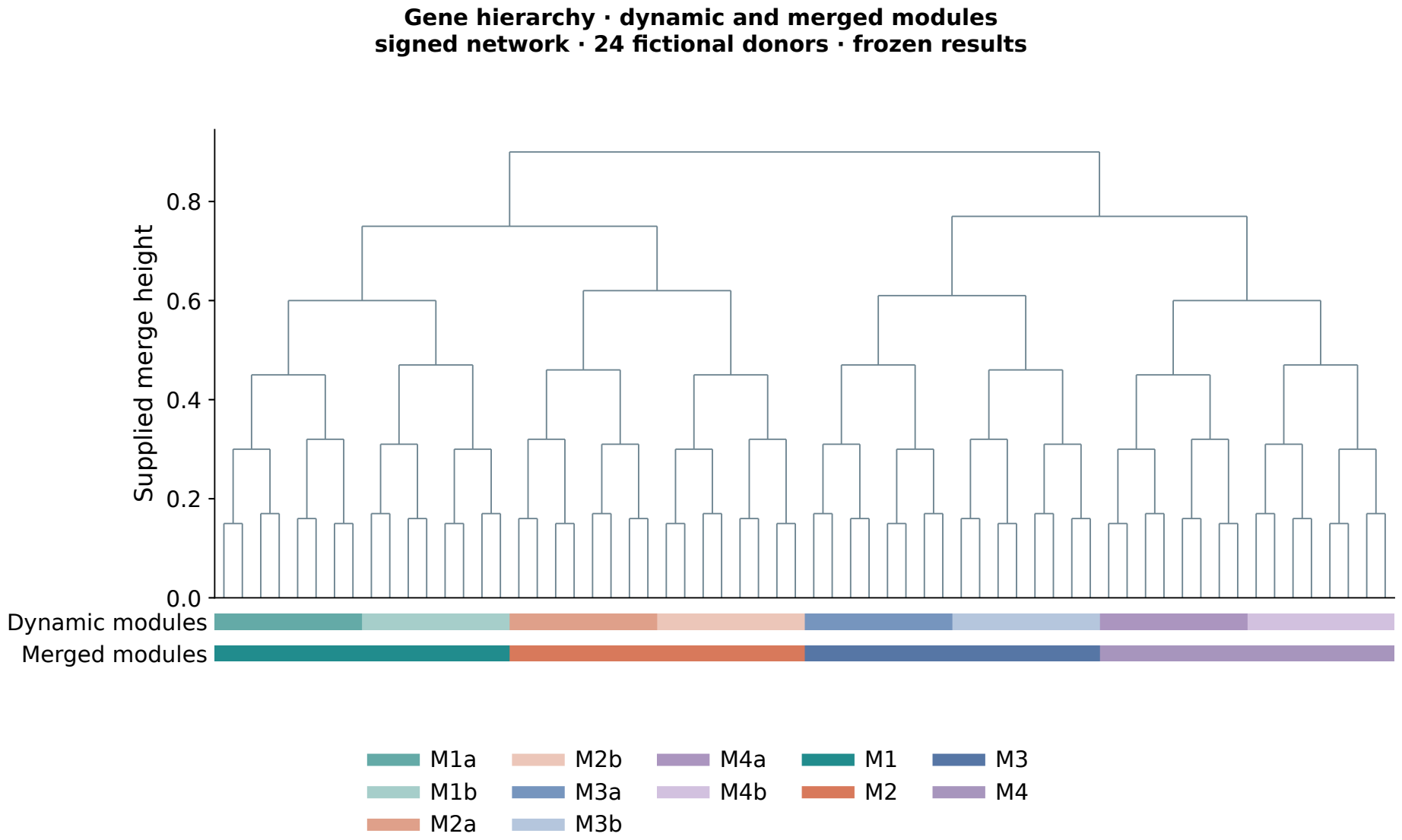


Soft power · topology and connectivity
signed network · 24 fictional donors · frozen results

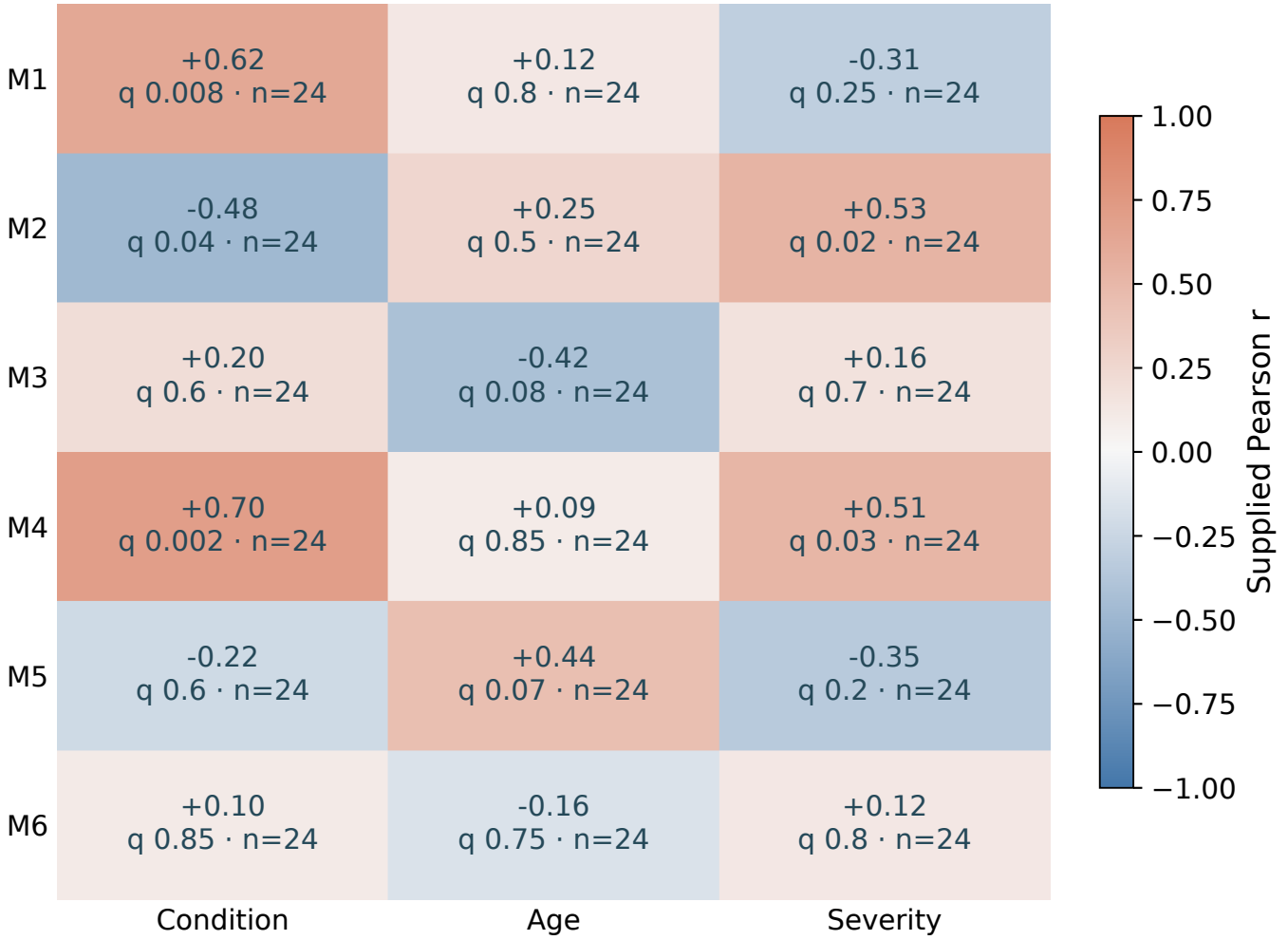


Samples · frozen clustering and design tracks
signed network · 16 fictional donors · frozen results

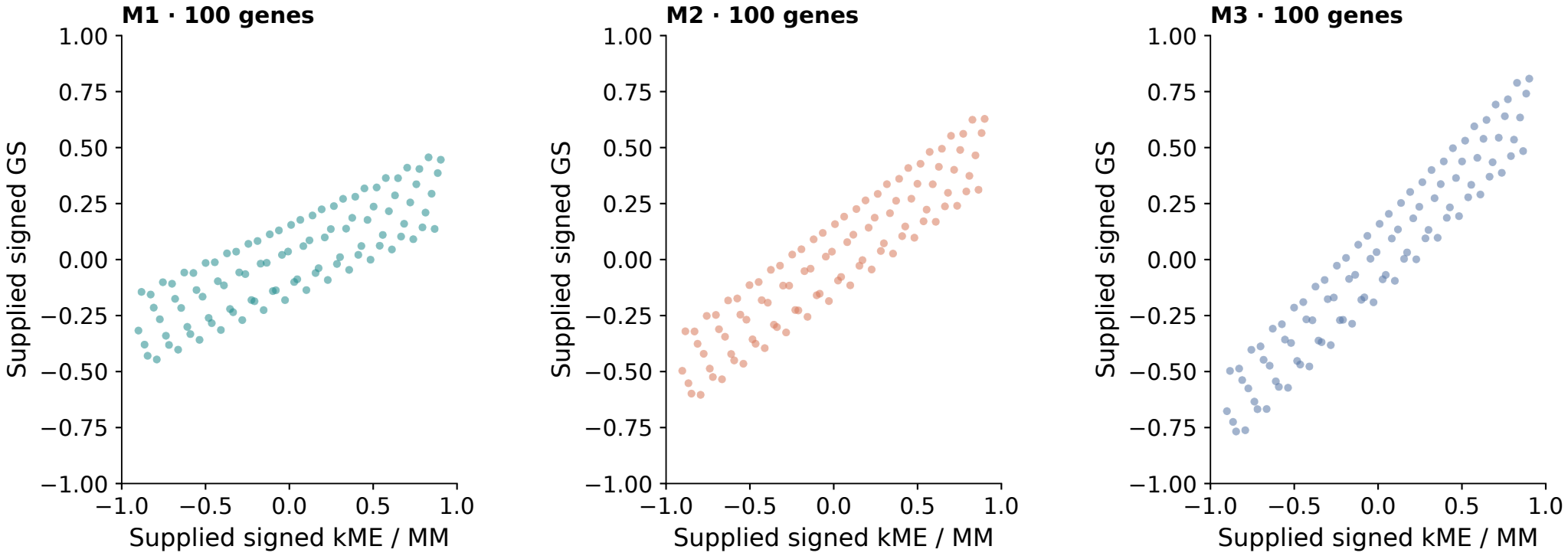



Gene hierarchy · dynamic and merged modules
signed network · 24 fictional donors · frozen results
Supplied merge height
0.0
0.2
0.4
0.6
0.8
Dynamic modules
Merged modules
M1a
M1b
M2a
M2b
M3a
M3b
M4a
M4b
M1
M2
M3
M4

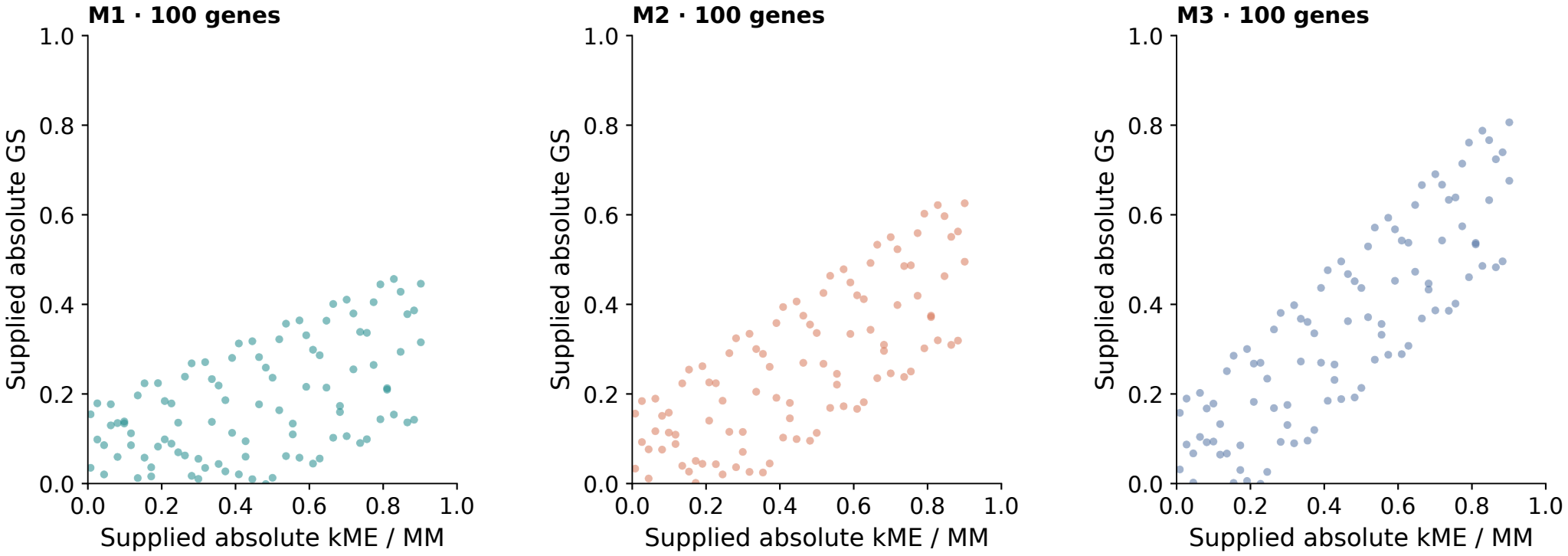
Module–trait associations · coefficient, q and effective n
signed network · 24 fictional donors · frozen results



Module membership and gene–trait association
signed network · 24 fictional donors · frozen results

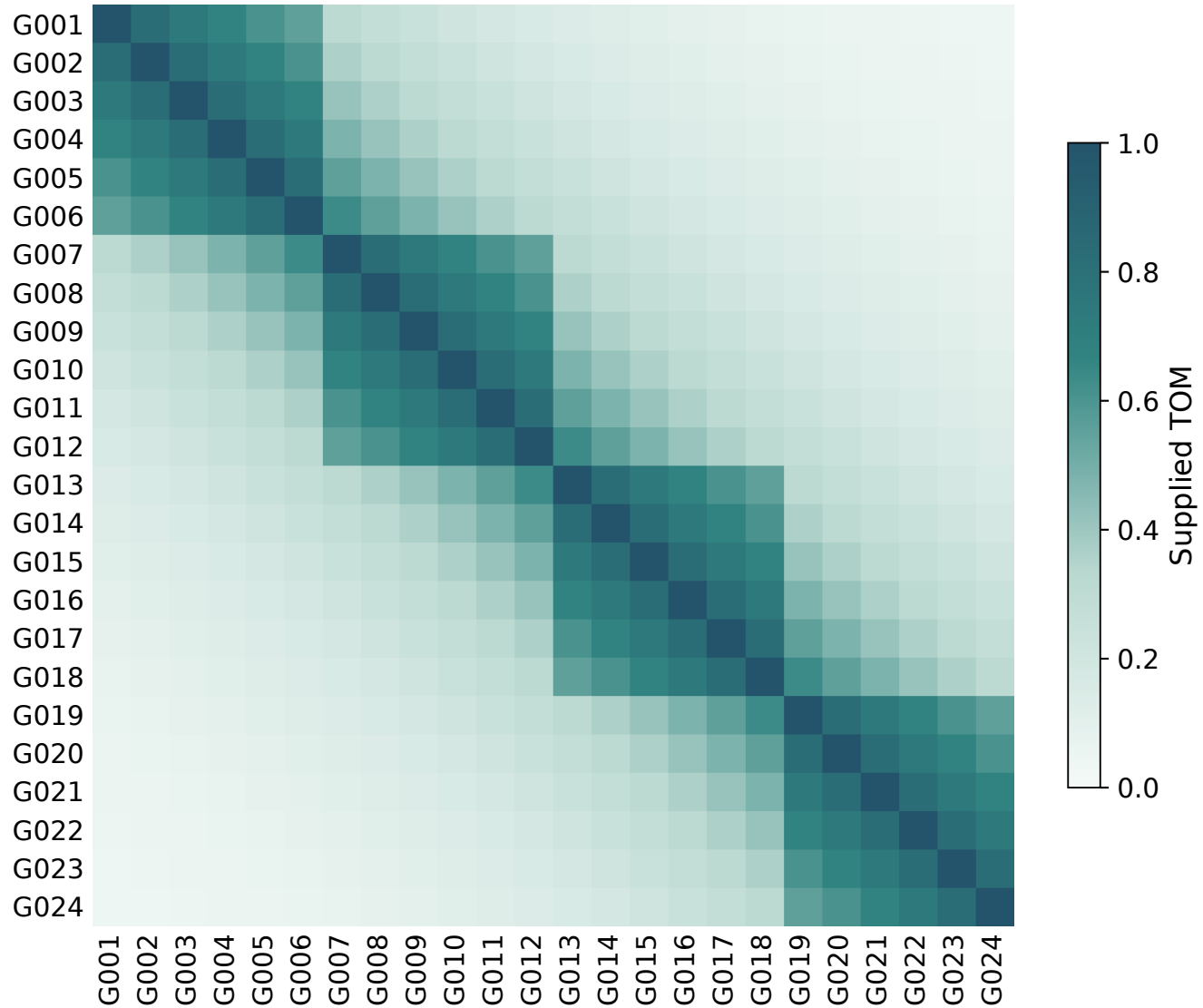


Upstream absolute MM and GS · separate declared convention
signed network · 24 fictional donors · frozen results

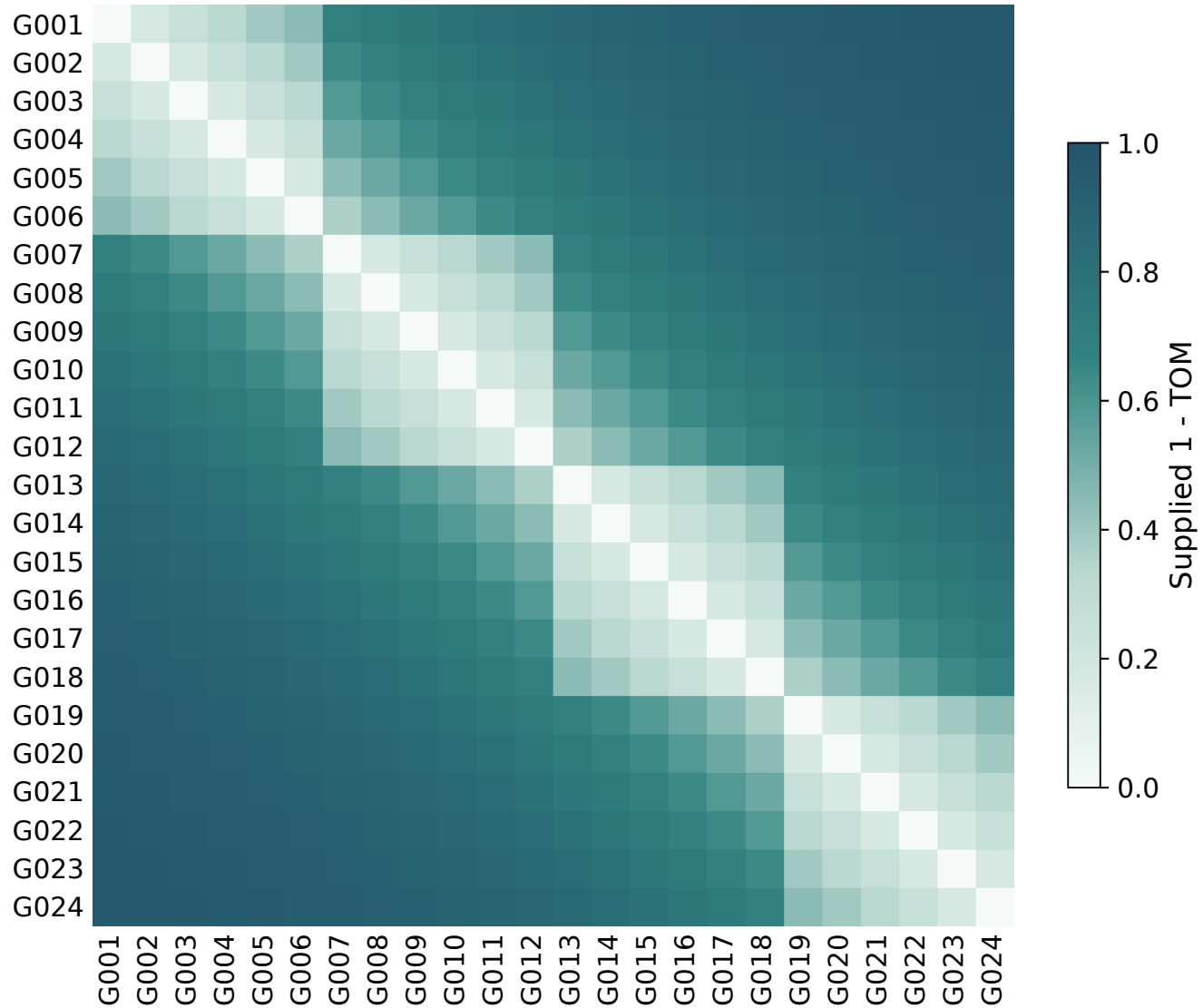


Software test fixture; not biological evidence

Topological overlap · complete selected frozen matrix
signed network · 24 fictional donors · frozen results



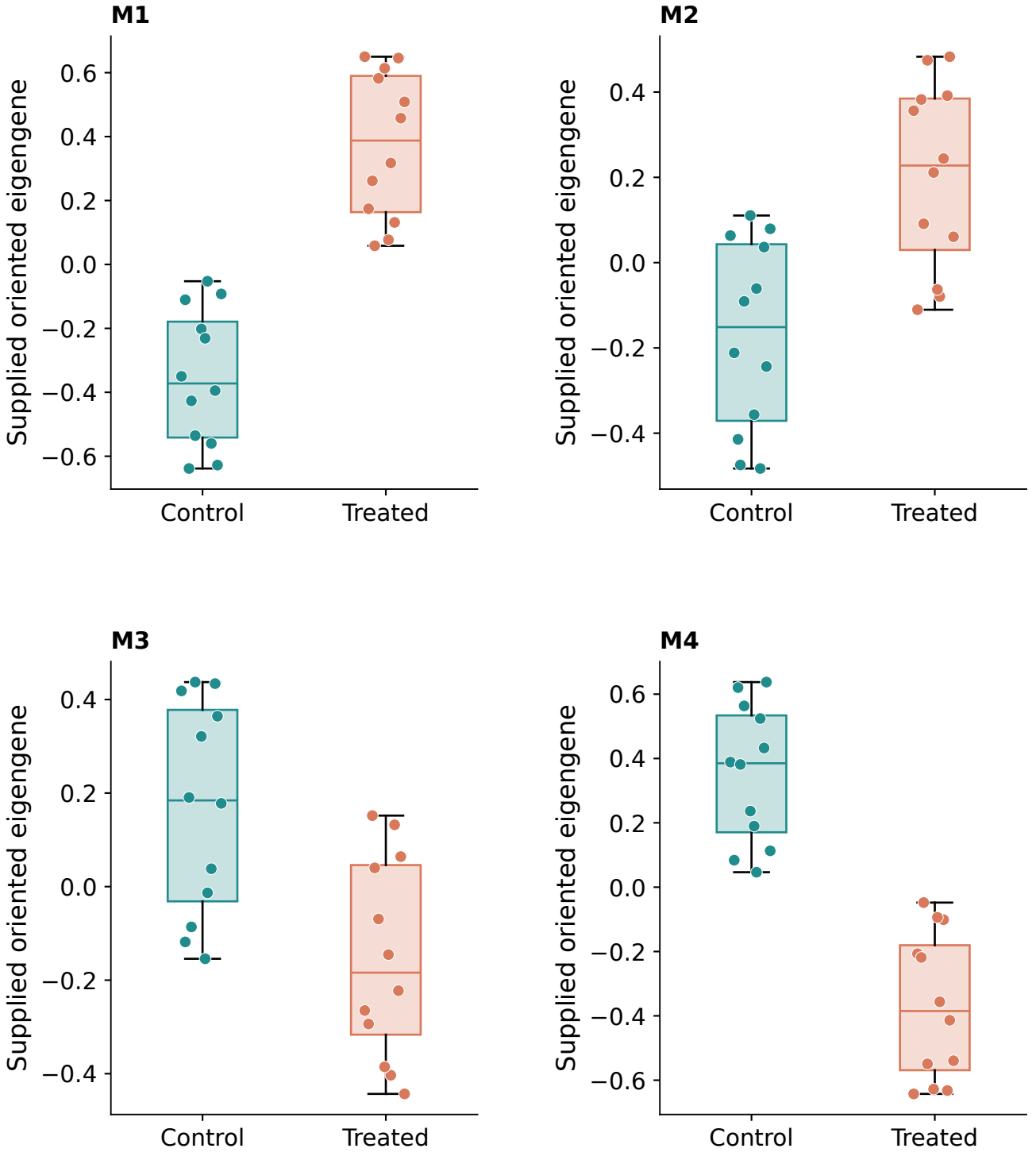
Topological dissimilarity · explicitly 1 − TOM
signed network · 24 fictional donors · frozen results

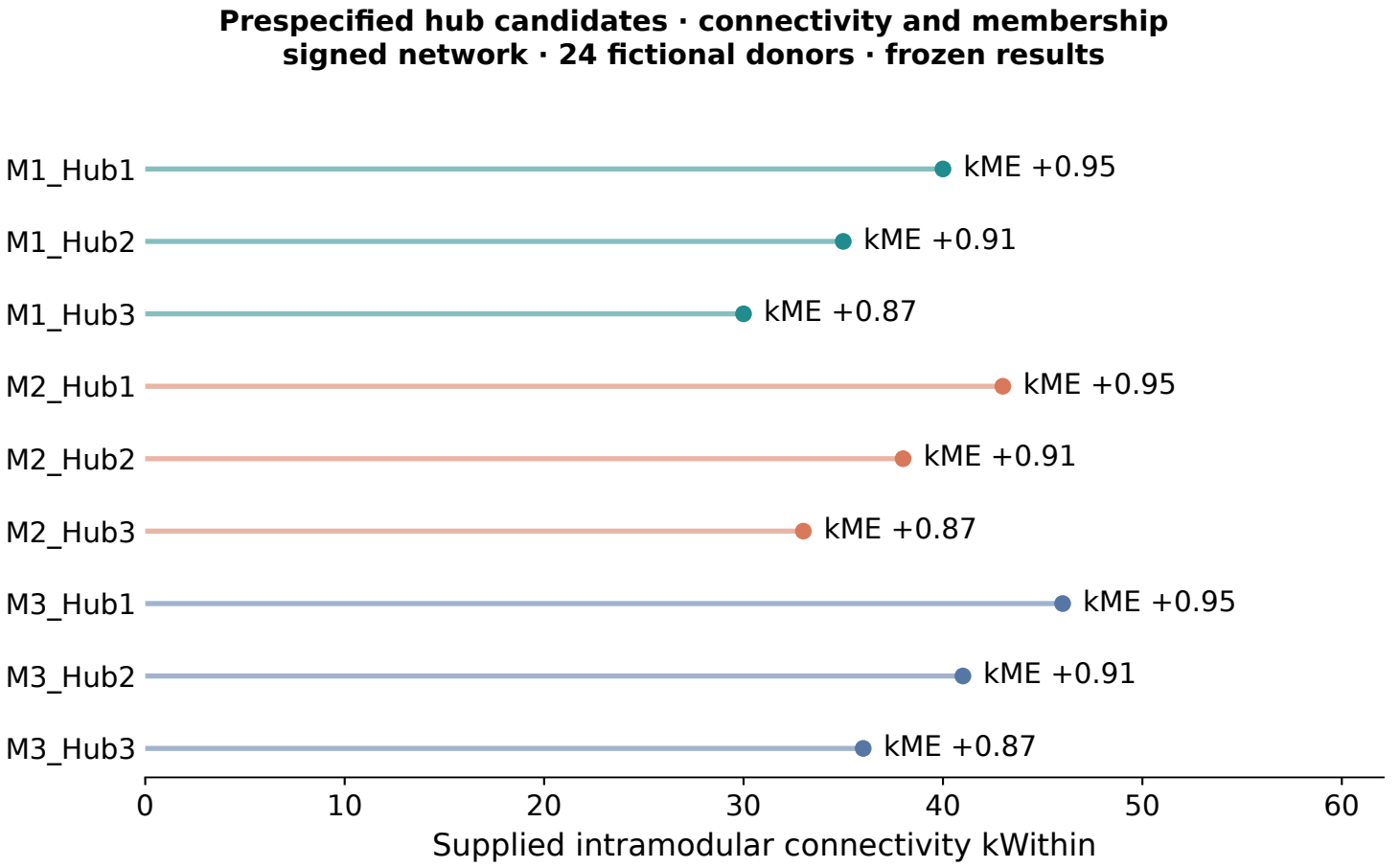


Relationships among oriented module eigengenes
signed network · 24 fictional donors · frozen results

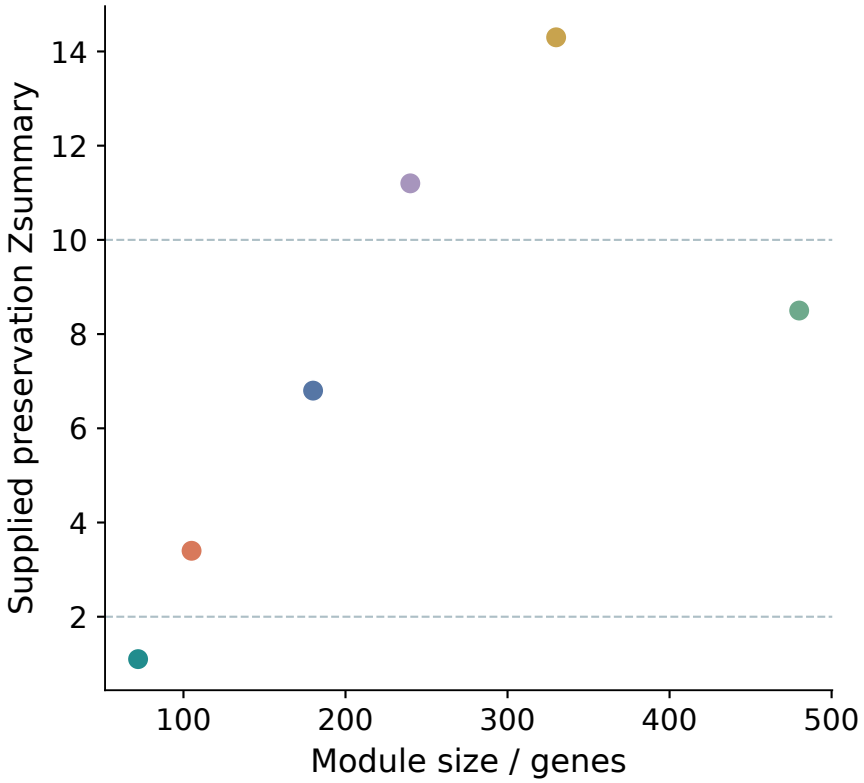


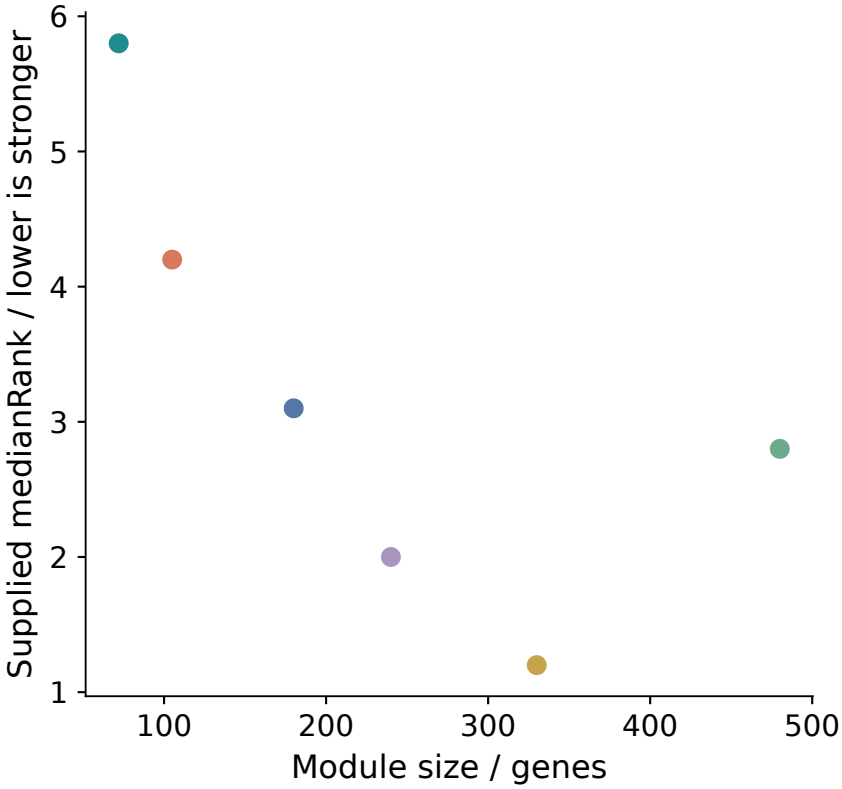
Donor-level module eigengenes · all units retained
signed network · 24 fictional donors · frozen results




Prespecified hub candidates · connectivity and membership
signed network · 24 fictional donors · frozen results
M1_Hub1
kME +0.95
M1_Hub2
kME +0.91
M1_Hub3
kME +0.87
M2_Hub1
kME +0.95
M2_Hub2
kME +0.91
M2_Hub3
kME +0.87
M3_Hub1
kME +0.95
M3_Hub2
kME +0.91
M3_Hub3
kME +0.87
0
10
20
30
40
50
60
Supplied intramodular connectivity kWithin

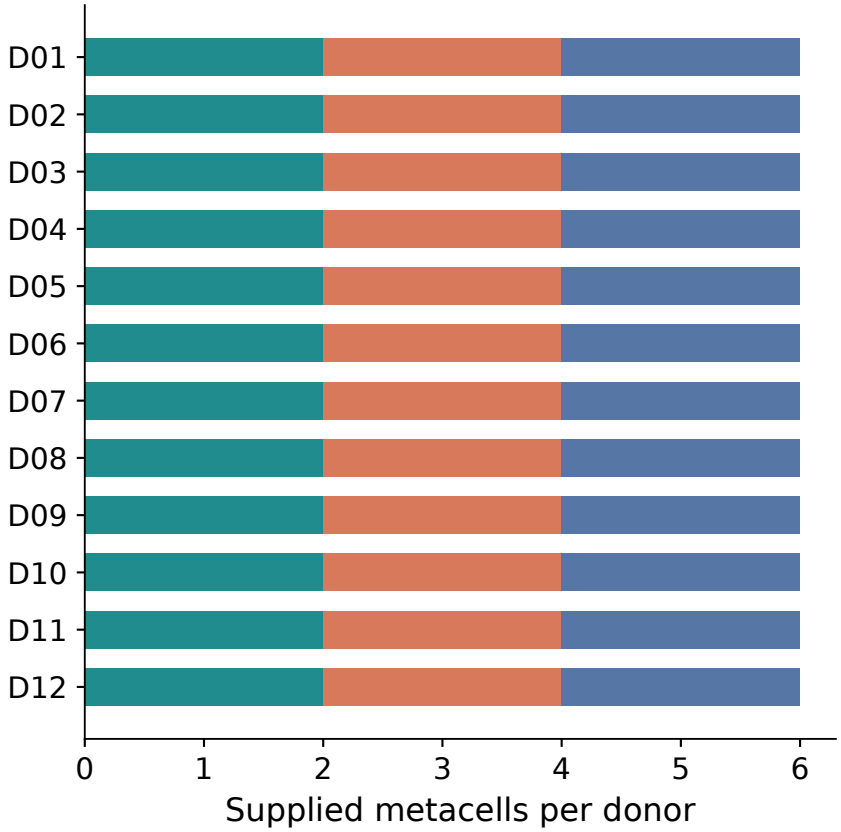
External module preservation · size, Zsummary and medianRank
signed network · 24 fictional donors · frozen results

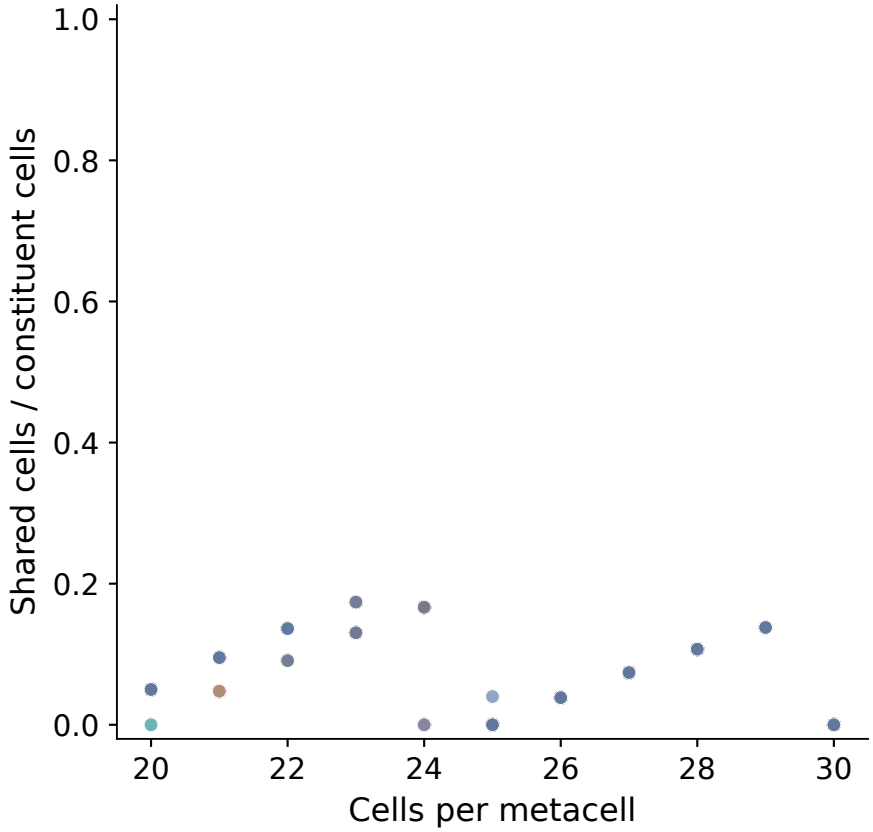




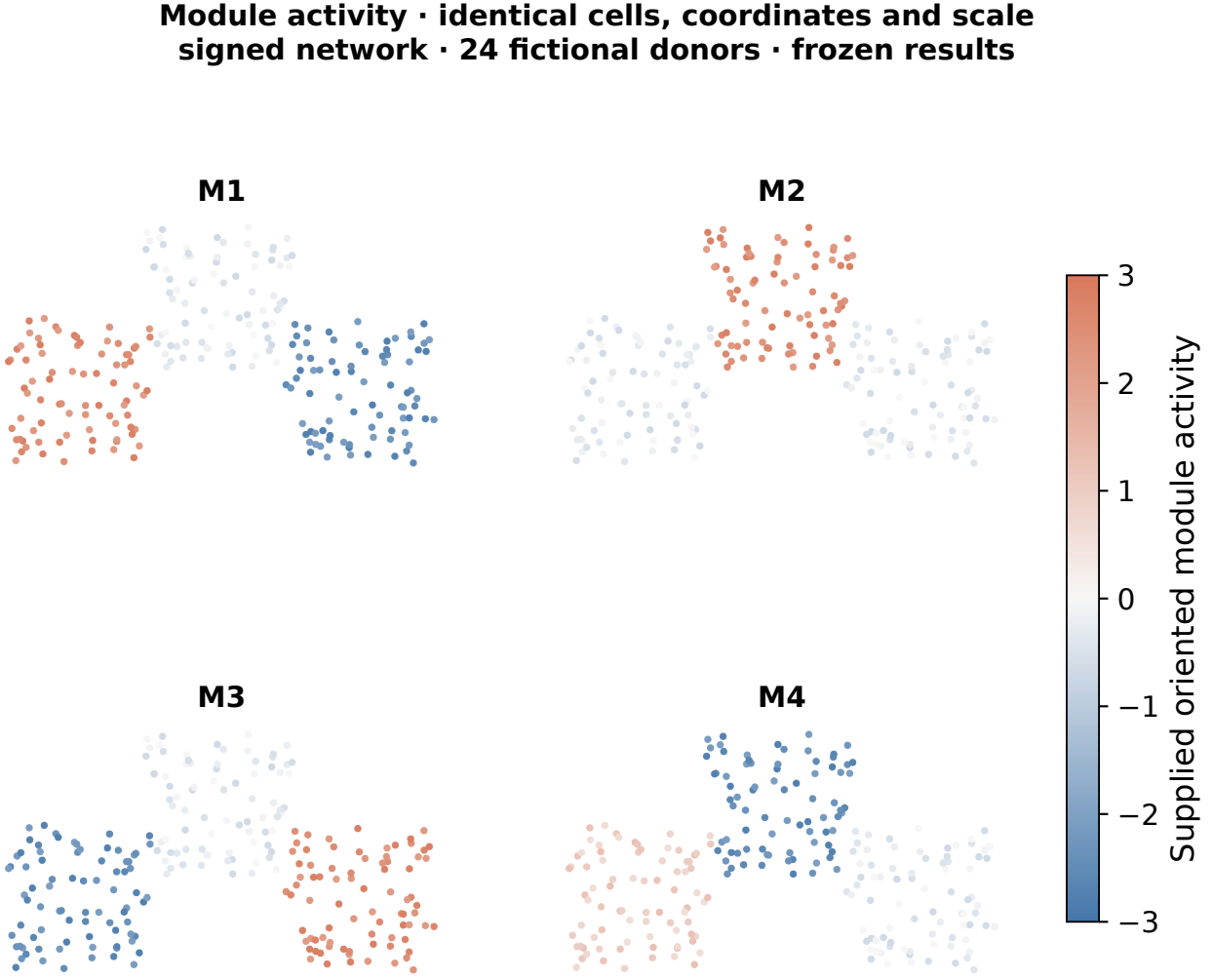
M1 M2 M3 M4 M5 M6

Metacell support · donors, celltypes and overlap
signed network · 12 fictional donors · frozen results

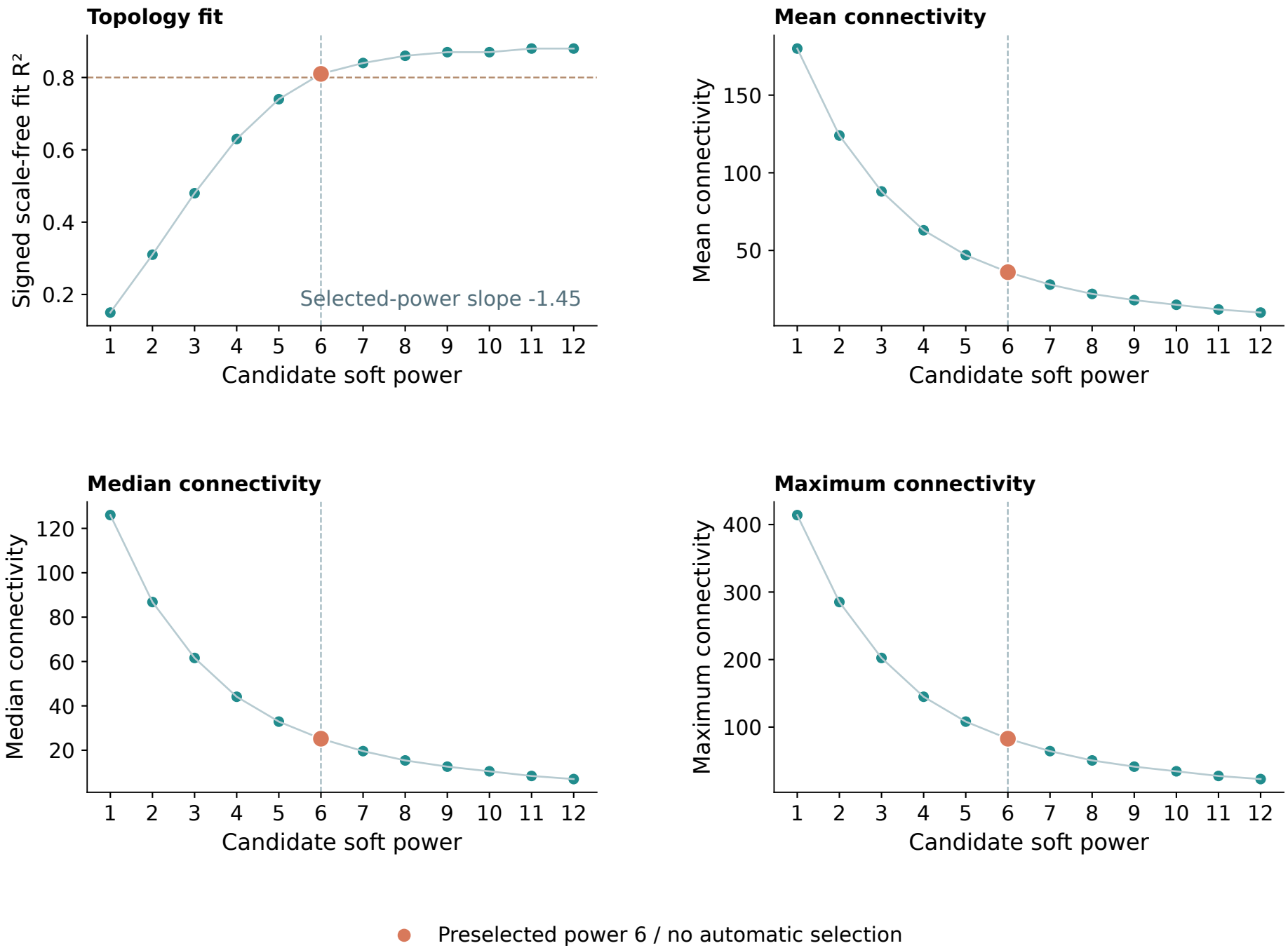


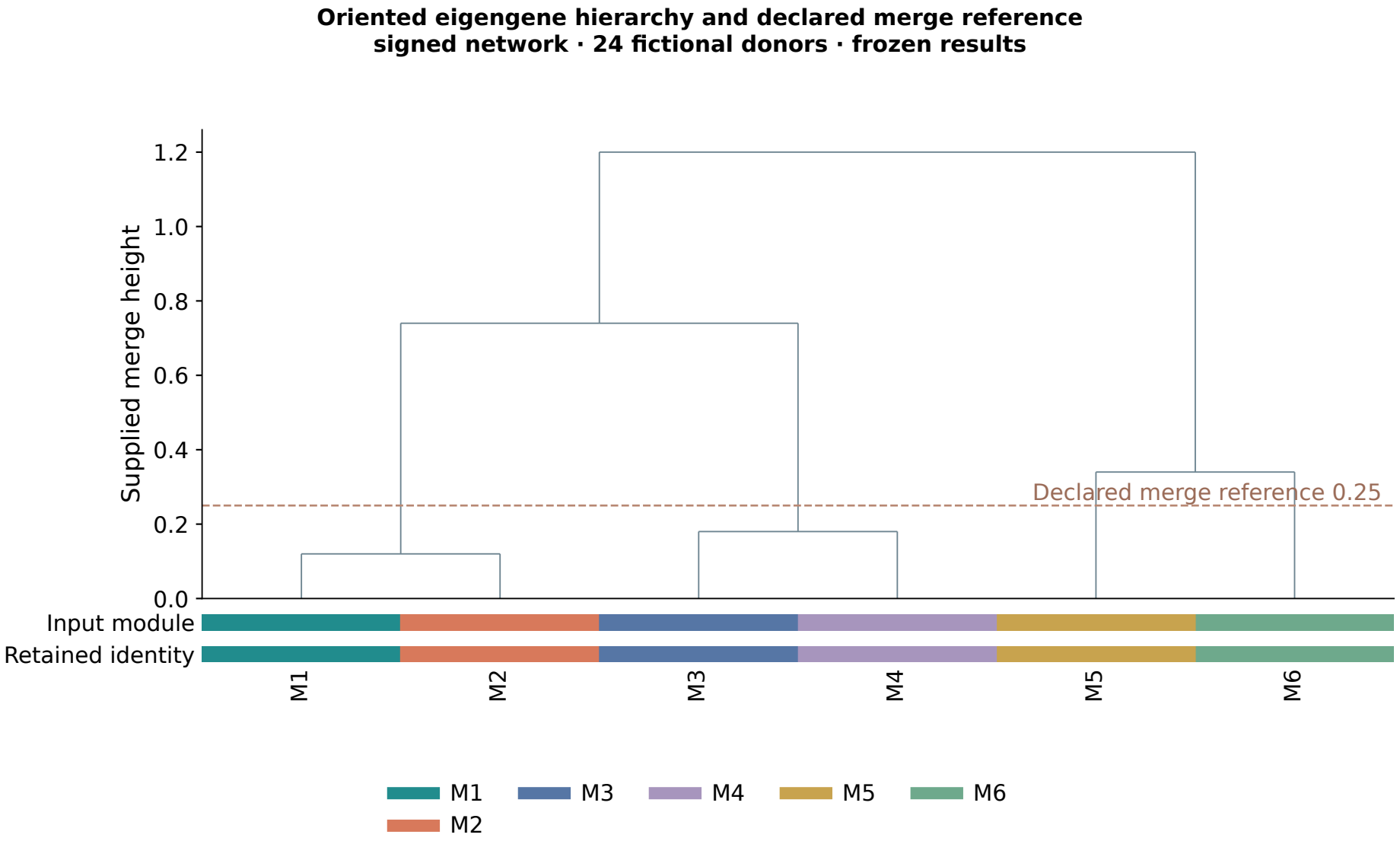


T cells B cells Myeloid

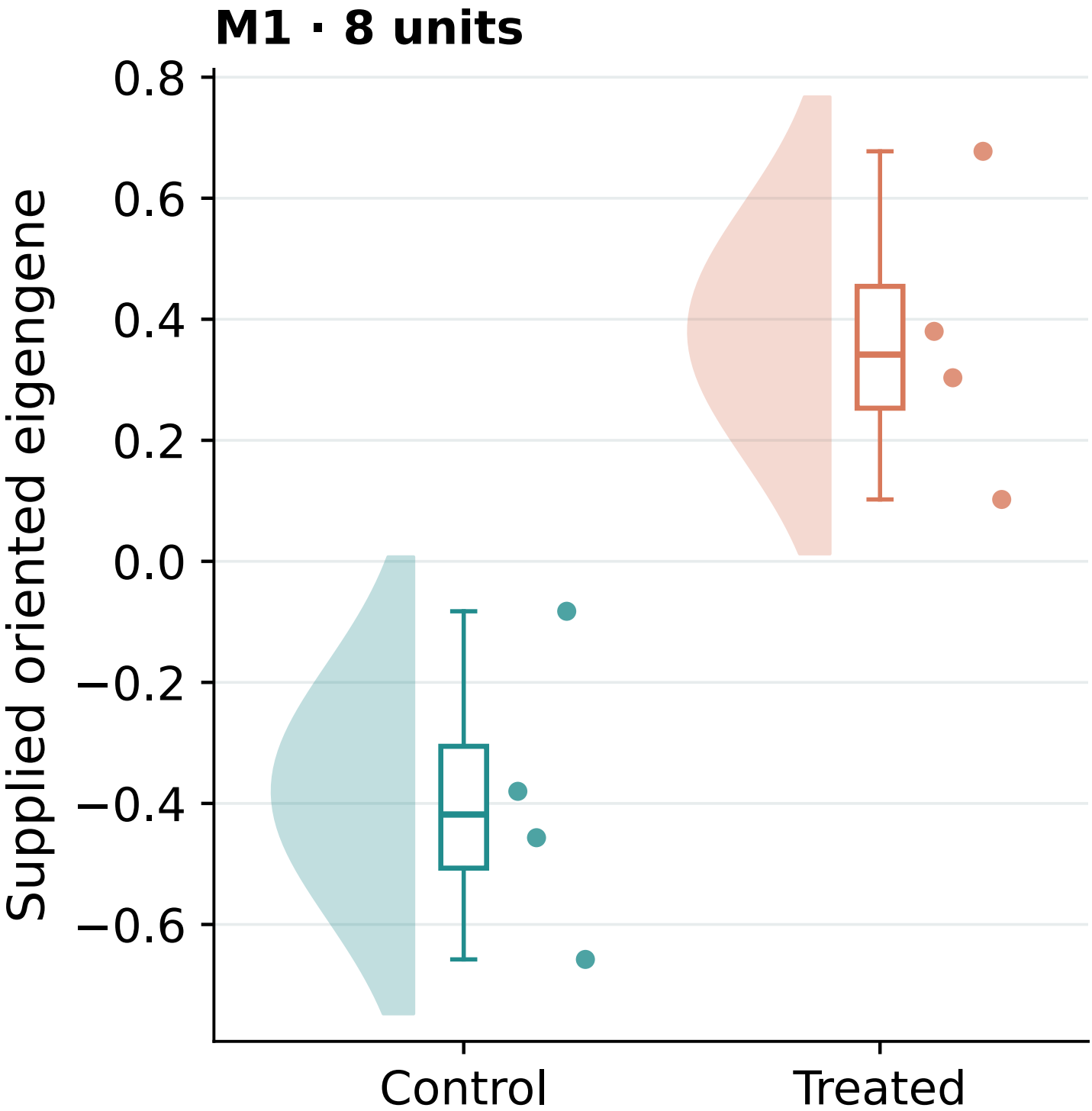

Module activity · identical cells, coordinates and scale
signed network · 24 fictional donors · frozen results
M1
M2
M3
M4
3
2
1
0
−1
−2
−3
Supplied oriented module activity

Soft-power diagnostics across all supplied candidates
signed network · 24 fictional donors · frozen results

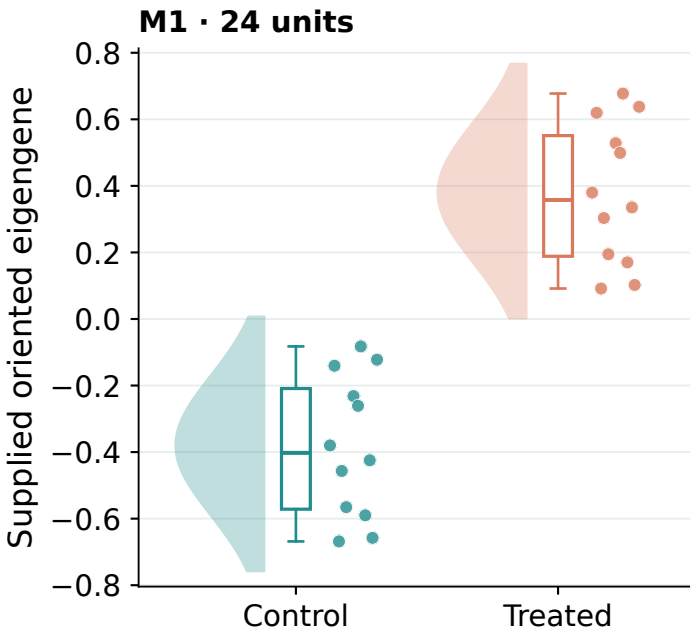


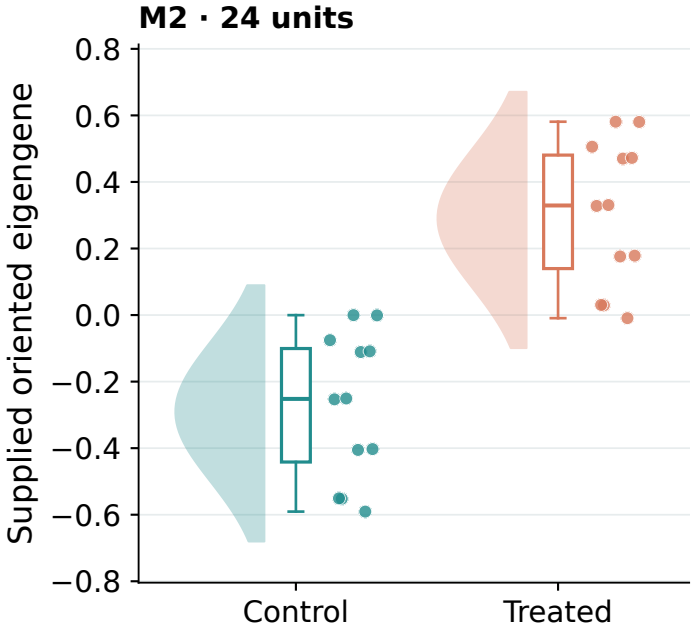

Oriented eigengene hierarchy and declared merge reference
signed network · 24 fictional donors · frozen results
Supplied merge height
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Declared merge reference 0.25
Input module
Retained identity
M1
M2
M3
M4
M5
M6
M1
M3
M4
M5
M6
M2

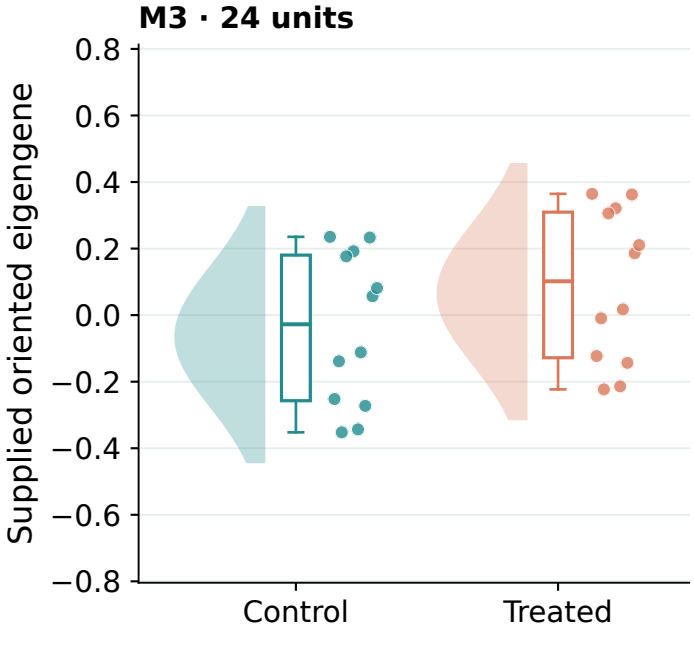
Eigengene distributions · 8 donors

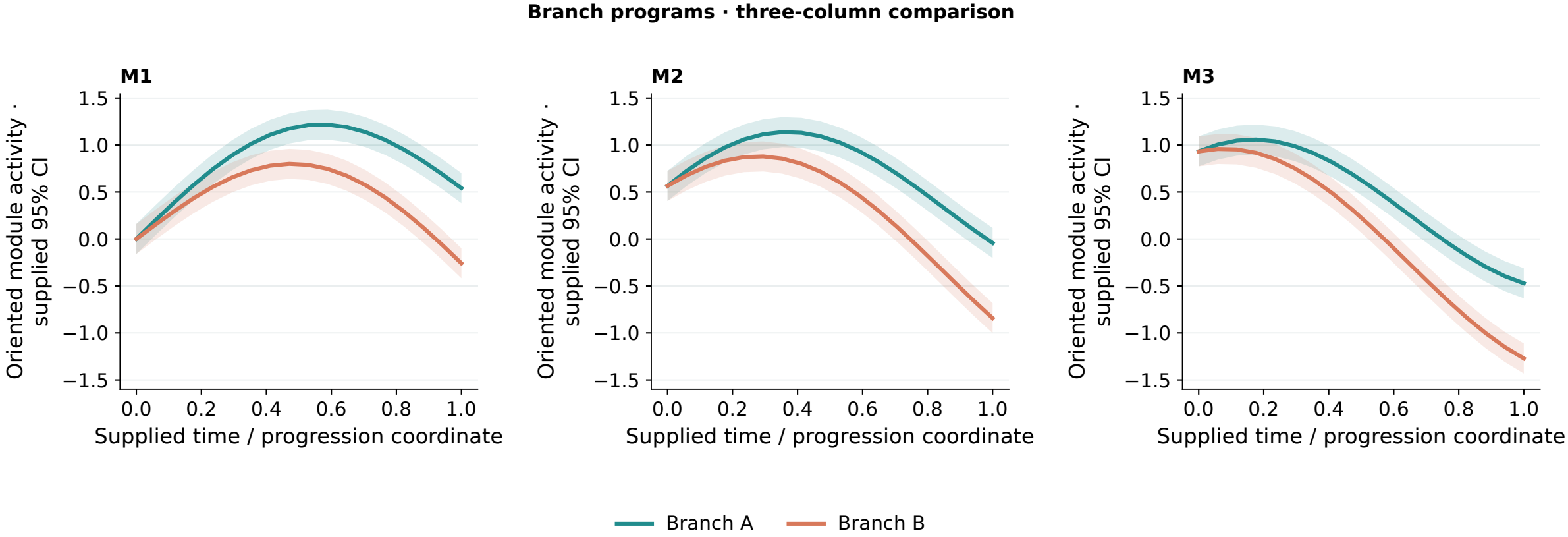


Eigengene distributions · one-column reading

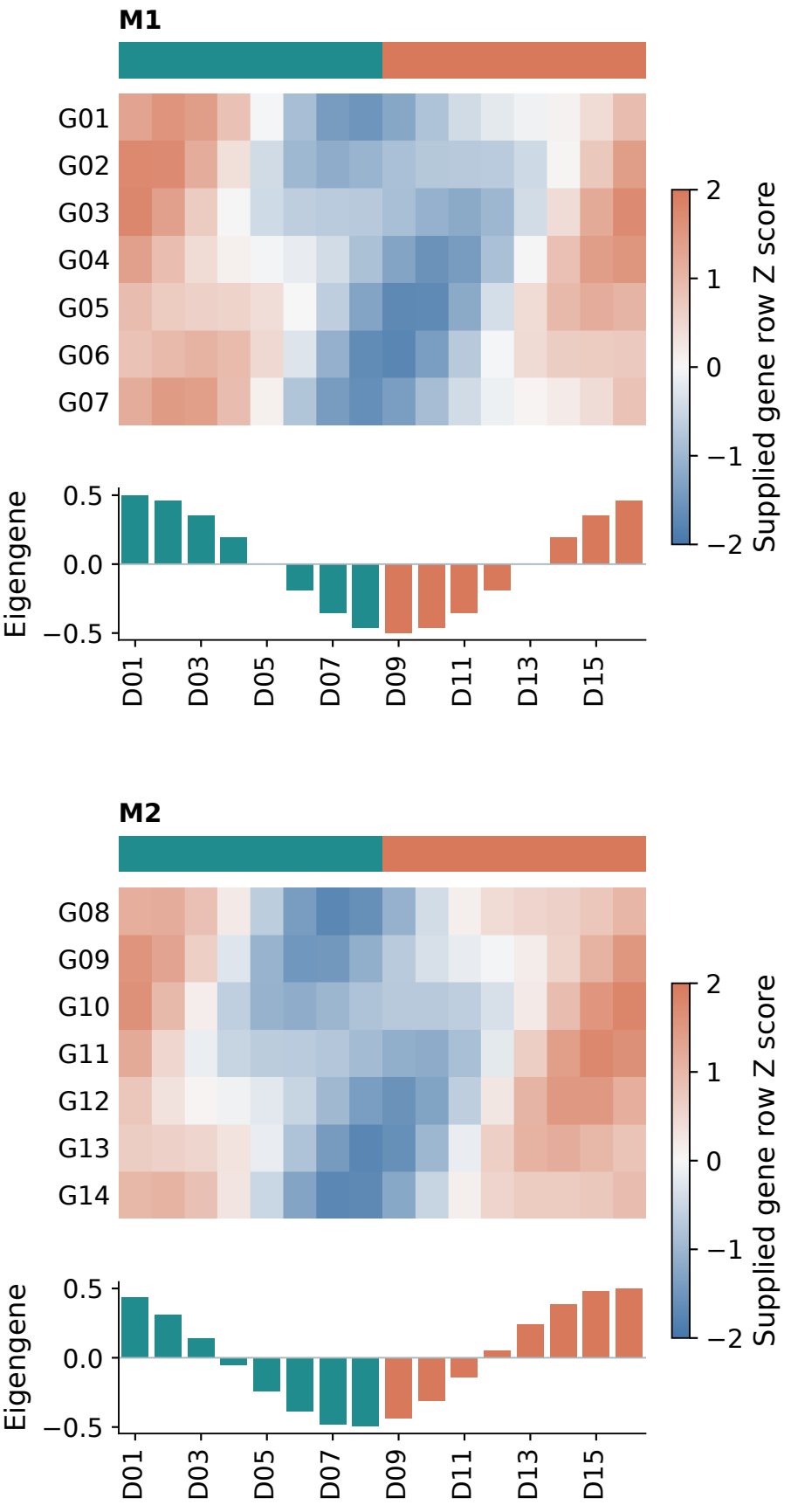




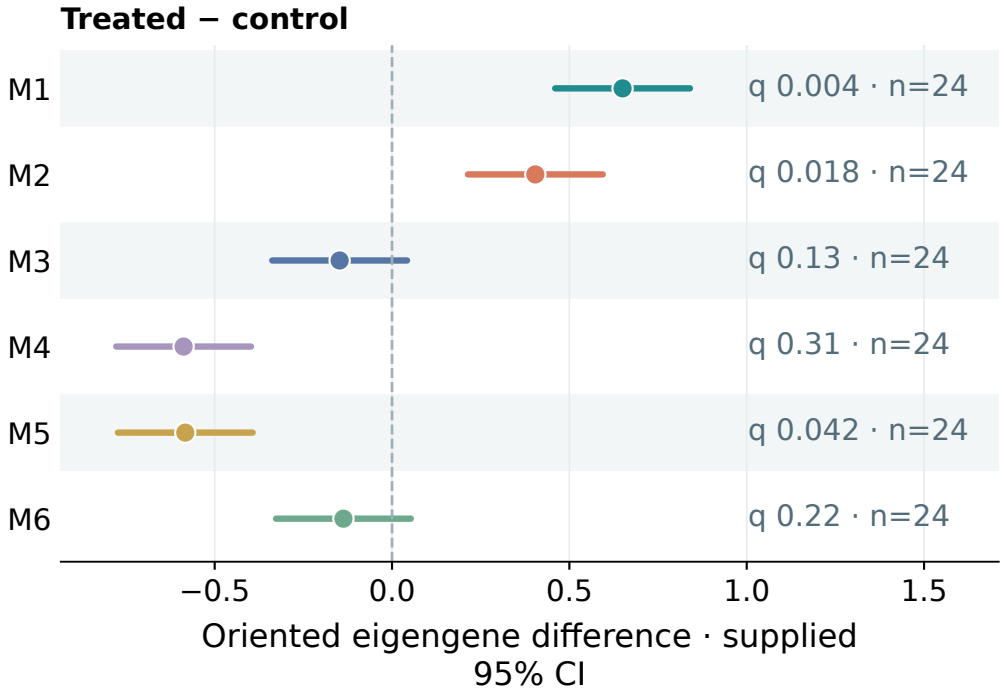


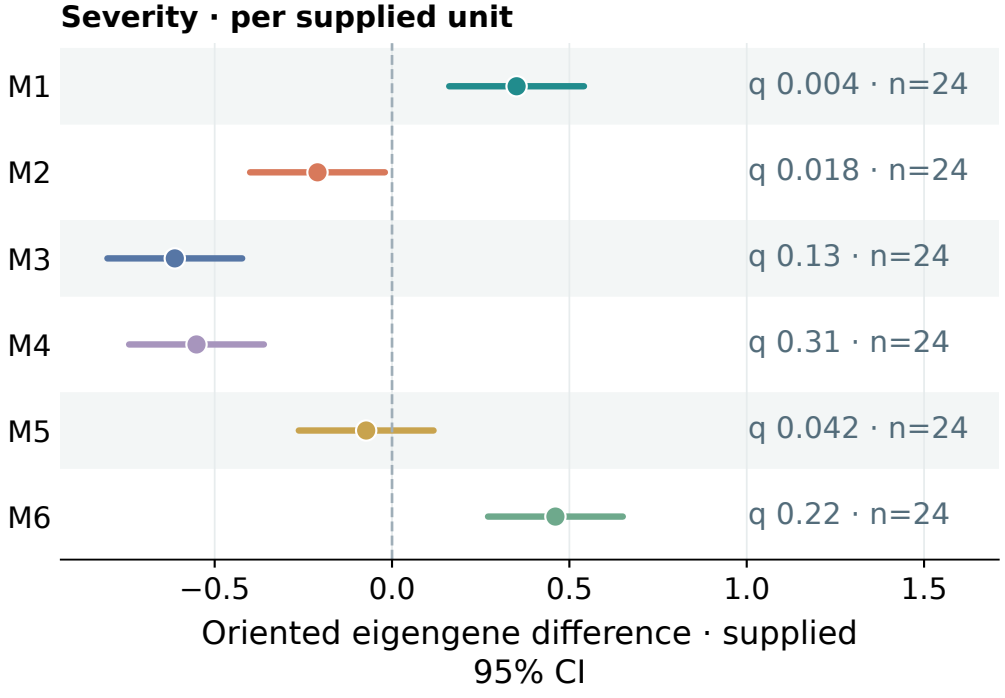

Branch programs · three-column comparison
M1
M2
M3
Oriented module activity · supplied 95% CI
Supplied time / progression coordinate
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0.0
0.2
0.4
0.6
0.8
1.0
Branch A
Branch B

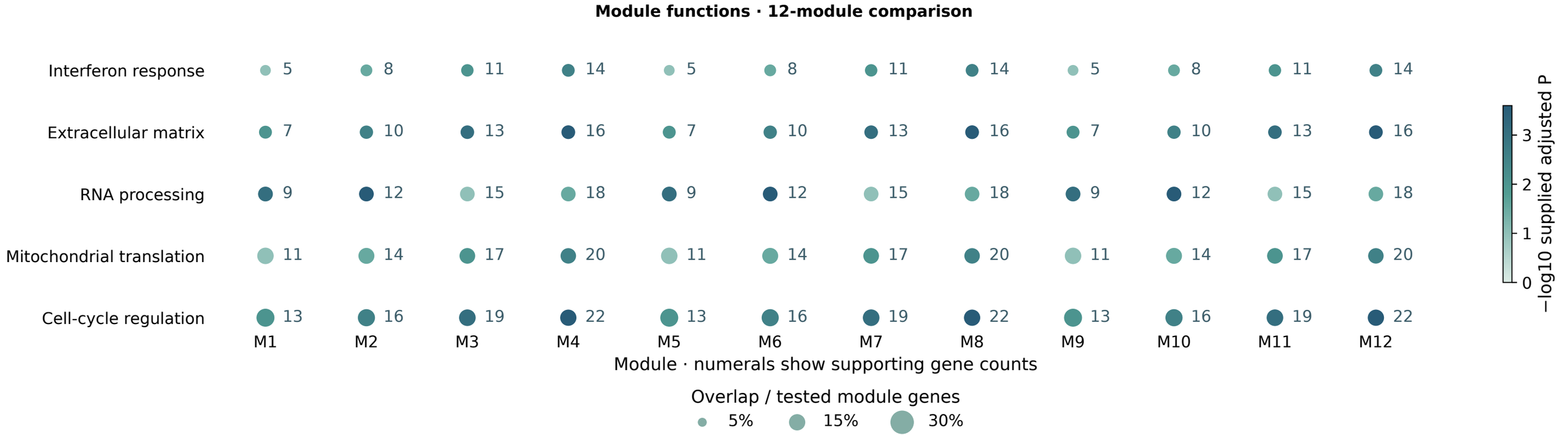
Module expression · one-column reading



Module effects · one-column reading







Software test fixture; not biological evidence